Preface

Pharmaceutical organic chemistry is the main branch of organic chemistry deals with the study of preparation, structure and reactions of organic compounds. As it deals with all the chemical reactions related to life, study of Pharmaceutical organic chemistry is important. Application of Organic chemistry in the development of pharmaceuticals, resulted in evolving Pharmaceutical organic chemistry. Hence studying Organic chemistry and applying this knowledge in Pharmaceutical substances is called as Pharmaceutical organic chemistry. Organic chemistry forms the basis of biochemistry, in which various aspects of health and diseases are studied. The biochemical knowledge is very important for the practice of nutritional, medical and related life sciences. In addition Organic chemistry paved way for the development of medicinal chemistry, Pharmaceutical organic chemistry, bioinformatics, biotechnology, gene therapy, Pharmacology, pathology, chemical engineering, dental science and so on. Organic substances play such a vital role in our daily life that all of us should know about organic chemistry in order to understand the manner how it influence our life process.

In the given conditions, a specific compound is inert or reactive, and if it is reactive, how will it react? Such knowledge will be used to design the structure of a substance that will have a specifically desired property. We will then know what substance to be used for making the parts for various instruments? Which drug to be used for a specific disease? Many of careers such as Doctors, Engineers, Pharmacists, Veterinarians, Dentists, Pharmacologists and Chemists make use of the knowledge of this fundamental subject. Hence the study of this basic Organic chemistry subject will make you to become successful professionals.

The field of organic chemistry is incredibly vast subject. Why is organic chemistry considered to be so difficult? in which "How to start" ? "What to study" ? and "How to study and remember the chemical reactions"? Is this either an antiquated misconception, or is absolutely true ? ? ?

The book has been designed to meet the needs of the intended readers with reference to the above questions.

Students should start learning organic chemistry by understanding only, not through mechanical memorization like "a poem learnt by rote in childhood". Organic chemistry is not a difficult subject, and once you understand, it will become an enjoyable subject and you blast your way by proposing your own way of reaction one after another.

This book is a product of my vision to design the best book on Pharmaceutical organic chemistry, which deals with the origin of organic chemistry, the concise description of structure of atom & organic molecules and their related properties, the nature of organic reactions & their mechanisms, nomenclature of organic compounds, clear classification of various organic compounds, preparation of each class of organic compounds by various routes, its chemical structure, physical properties and chemical reactions with the mechanism in a simplified manner and drugs derived from each class along with their applications in medicine. Swathing the entire features of Pharmaceutical organic chemistry, first of its kind, is the unique feature of this book. It facilitates the students to understand the subject more easily and make the subject interest.

As the students entering graduate course, understanding of Pharmaceutical organic chemistry is always been a difficult task especially the various chemical reactions with their mechanisms of different kind of organic compounds. Hence my efforts have been devoted to authoring a book by equipping with the challenging requirements of the subject for the new generations of the teachers all over India and is easy to read for students who are not necessarily of Pharmacy program, but mainly for the students of first time reading organic reactions, their mechanisms and their applications in science, Pharmacy and medicine.

Methodical description of each chapter, enriching chemical and pharmaceutical background of the medicinally important organic compounds, proceeding each group of organic compounds in a systematic way in easy-to-understand style in the larger interest of the students without any difficulty and making the reader acquainted thoroughly with chemistry of organic compounds is the unique feature of this book.

In preparing this text book, I have tried to enrich the importance of organic compounds in medicine and pharmacy, so that the anticipated audience of this book will feel the importance of organic compounds and made the book a comprehensive. The students of Pharmacy graduates of our country faced with the scarcity of books to serve their needs. Few of the authors dealt well about the basics of organic chemistry, but the reactions of organic compounds are not presented in a easy to understand manner which students felt difficult to learn. Some of the books of organic chemistry fail to give the chemical structures for all reactions, hence the students are unable to understand the reactions clearly. Hence the content of this book is made as a humble attempt to cater the needs of academicians belonging to all Indian Universities by incorporating the chemical structure of all reactions and enriching basic principles of each organic class of compounds.

The book has covered the entire Pharmaceutical organic chemistry, starting from origin of organic chemistry to advanced topics like stereochemistry and heterocyclic compounds and it is divided into 31 chapters. **Chapter 1 to 7** deals with the basics of the organic chemistry, wherein the fundamentals like origin and development of organic chemistry, structure of organic molecules and their related properties are described. Classification and nomenclature of organic compounds and general terms used are also presented in a systematic way, which is easy to understood and able to reproduce well in examinations.

Chapter 8 to 25 deals with Aliphatic and Aromatic compounds which are further divided into different chapters and each chapter is dealt with Introduction, Importance of each class of compounds, Nomenclature, General methods of preparation, General physical and chemical properties and Pharmaceutically important organic compounds of each chapter are described in a easy to understand manner. Summary of methods of preparation and chemical reactions in a flow chart manner presented is unique and helps student to remember for exam which is the first of its kind.

Chapter 26 to 28, Isomerism and detailed description of optical and geometrical isomerism is presented in a simplified manner.

Chapter 29 to 30, Heterocyclic rings of various types and their utility in pharmaceutical chemistry is described well. The medicinal compounds derived from each heterocycles also exemplified which makes the reader inspiring.

In **Chapter 31,** Important reactions and reagents used in organic chemistry and some of the special reactions are described with their mechanism and applications in Pharmaceutical organic chemistry.

To inspire the readers and make them attracted, interesting facts about great scientists and Organic compounds and their discovery etc are given under each chapters.

In order to help students to remember well and reproduce well in exams, the style and presentation are followed simple and similar pattern in all chapters. We have also used some special abbreviations like **"CAD"** (which helps to remember cold Alkaline dilute).

To help the students to learn and magnetize the attention we have used color in equations and diagrams. We have done it this thoughtfully and purposefully and not just to make the book attractive. These small inputs helps a lot for the fresher's who enter into degree program.

We hope that this special volume will be a good source of information and reference for not only to graduates and post-graduate students but also for basic and applied researchers in this field. Moreover it will also be of interest to a wide range of scientists who are involving in the Pharmaceutical organic chemistry related research. I welcome suggestions and constructive criticism from all corners of scientific community.

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What is the best way to study organic chemistry?

- Develop the desire to study organic chemistry
- Read the basic points thoroughly
- Read the chemical structures and correlate with the basics studied
- Be through with the nomenclature
- Always prepare your own notes by applying the above knowledge
- Read each and every step of the reaction and mechanism thoroughly
- Practice daily
- Make use of the lab time
- Discuss in a group and clear the doubts spontaneously
- Break large tasks in smaller ones

Acknowledgement

A warm response to my earlier Books on the "Text Book of Medicinal Chemistry", "Pharmaceutical Chemistry of Natural Products" (Published by Elsevier), Pharmaceutical Inorganic Chemistry (Published by Pharma Book Syndiacte) prompted me to write this Book on "Pharmaceutical Organic Chemistry", covering all the topics suggested by the Pharmacy Council of India (PCI) for graduate students. I dedicate this book to the many hundreds of budding graduates and pharmacy students, whom I have taught over the years and my Teachers who have encouraged me to convert my class notes into the text book in order to reach into wide range of academic community.

It is my pleasure to place on record my heartfelt thanks to everyone who have made this book possible, especially my beloved teachers of 10+2 class to Ph.D., who made me to read the Organic Chemistry in a simplified manner while the other students felt difficulty in reading and reproducing in the Exams.

I am immensely grateful to Prof. K. Chinnaswamy, Dr. B. Suresh (President, Pharmacy Council of India) and Dr. R.K. Goyal, Dr. Rajani Giridhar, Dr. M.R. Yadav, Dr. C.J. Shishoo and Dr. U.S. Pathak for their constant support, inspiration and initiation extended to me to author this book.

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I also express my sincere thanks to my father-in-law (Shri. V. Sundara Rajan) and mother-in-law (Ms. S. Chinnammal) for their kind encouragement and moral support throughout my Career. My father-in-law born in a small village, studied upto 10th class only, had joined as a Police and came up to the level of Police Inspector is a great motivation for me.

In all my academic efforts the stimulation I gain from my father (Mr. P. R. Veerachamy), mother (Ms. V. Kamuthai), sister, brothers and wife A. Sathyabhama to reach this goal is like the nature provides sunlight for photosynthesis for healthy maintenance of humans, and the patience and cooperation extended by my children, A. Dharshini Aishwarya (B.Pharm) and A. Dharshini Abhinaya made me think of the goal without any diversion. To express my thankfulness, I pray the Almighty to bless my children with teachers like those I got in my life so that they too are inspired by their teachers and dedicate to the field of Pharmacy and, in turn, serve for the mankind.

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2

Nomenclature of Organic Compounds

Introduction

In early days, scientists named the compounds based on historic background. For example, "wood spirit" was named so because it was obtained from distillation of wood. Later it was named as methanol based on Greek words (methu = wine and hale = wood). Similarly, the name "acetic acid" was derived from vinegar (Latin; acetum = vinegar), because the acetic acid is the major constituent of vinegar. These are called as common names or trivial names.

As the number of compounds were discovered more, naming by history becomes difficult. Hence systematic naming becomes important. The systematization of names was carried out by the International congress of leading chemistry held in Geneva, 1892.

Rational system of nomenclature was formed and it is called as Geneva system of nomenclature. Slight revision and improvements were carried out time to time. One such being held at Liege (Belgium) 1930 by International Union of Chemistry and it is called as IUC system of nomenclature. The IUC was later modified by the International Union of Pure and Applied Chemistry in 1958 and it is called as IUPAC system of nomenclature.

To name the organic compound according to IUPAC nomenclature a set of rules were framed and all the compounds are named accordingly. However, even today some of the common names are used for organic compounds. Hence the chemists should also be aware of the common names apart from IUPAC nomenclature.

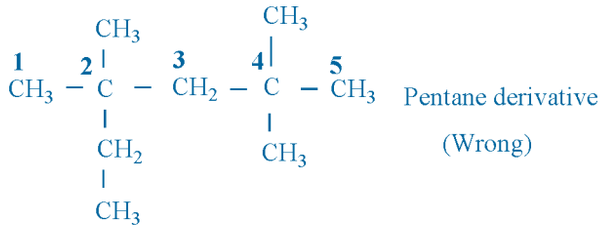
Non-systemic nomenclature of organic compounds like common name, trivial name *etc* are described in individual chapters.

IUPAC System of Nomenclature

IUPAC nomenclature is been used now a days to name organic compounds. However, some of the simple compounds are named by trivial names. Earlier names have been continued even today but complex organic compound can be given using IUPAC nomenclature only. Various rules are followed for naming compounds by IUPAC system:

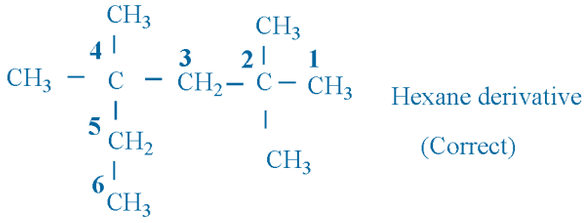
Rule 1: Longest chain rule: "In the given organic compound longest possible chain of carbon atoms is selected and the compound is named as a derivative of this alkane."

For example, the compound given below have five carbons in horizontal line and six carbons in the longest chain hence we should select it as a hexane derivative only.



Pentane derivative

(Wrong)

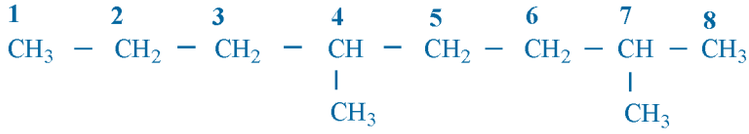


Hexane derivative

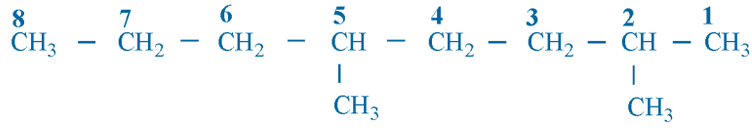
(Correct)

Rule 2: Lowest number for substituents rule: After selecting the longest chain, the numbering should be given from one end to the other end. While giving the number, the substituents should be given lowest possible number.

For example, the compound given below is named in two ways.



4,7-Dimethyl octane (Wrong)

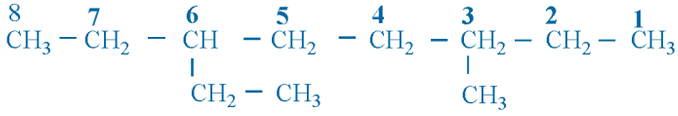


2,5-Dimethyl octane (Correct)

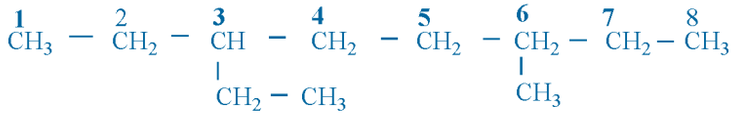
In first case, naming 4,7-dimethyl octane is not correct because 2,5-di methyl octane have lowest numbers for the substituents.

If different alkyl groups are in equivalent positions in relation to the end of the chain, preference is given to the end where the radical has fewer carbon atoms (methyl, ethyl, *etc*).

In the following example, the first case of naming is correct because methyl group is given preference over ethyl group.



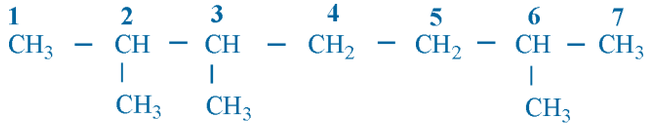
6-Ethyl-3-methyl octane (Correct)



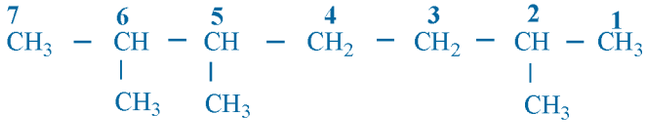
3-Ethyl-6-methyloctance (Wrong)

If identical radicals are at equal distance in the chain then the numbering starts from the end where it is more branched.

In the following example, the first way of naming is correct where the branching end is given preference.



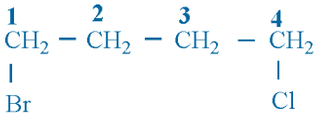
2,3,6-Trimethyl heptane (Correct)



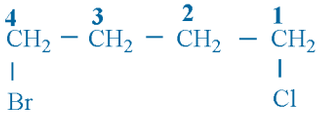
2,5,6-Trimethyl heptane (Wrong)

If two sets of numbers are possible for the given chain, then order of prefix in the name will decide the numbering (alphabetical order of the substituents).

For example, the given compound can be named as 1-bromo-4-chloro butane or 1-chloro-4-bromo butane. As the prefix bromo is first, the first name is correct.



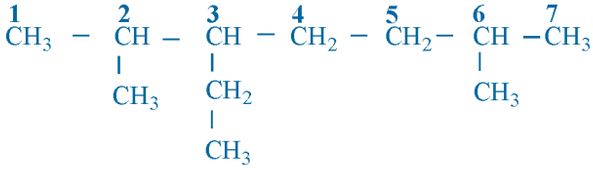
1-Bromo-4-chloro butane (Correct)



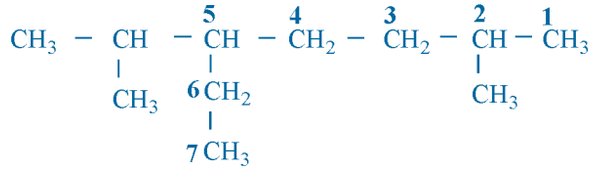
1-Chloro-4-bromo butane (Wrong)

If chains of equal length are competing for selection as the parent chain in a branched alkane, the preference goes to the chain carrying more branches.

For example, in the given organic compound first way of naming *i.e.,* 3-ethyl-2,6-dimethyl heptane is correct where as 5-isopropyl-2-methyl heptane is wrong.



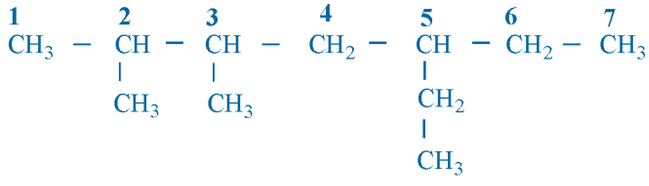
3-Ethyl-2,6-dimethyl heptane (Correct)



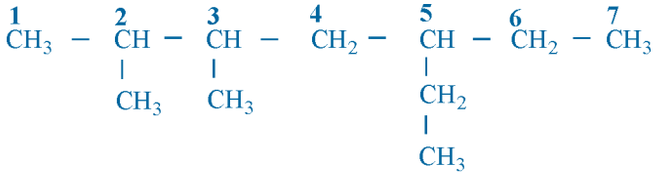
5-Isopropyl-2-methyl heptane (Wrong)

Rule 3: Arrangement of prefixes: When there is more than one group attached in the chain, they should be arranged alphabetically. If same group is presented in two or three places of chain then the prefix di or tri *etc* are used.

For example, the given organic compound is named as 5-ethyl-2,3-dimethyl heptane.

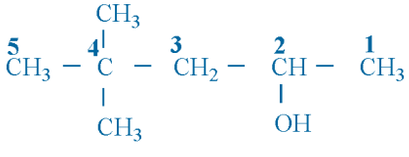


5-Ethyl-2,3-dimethyl heptane (correct)

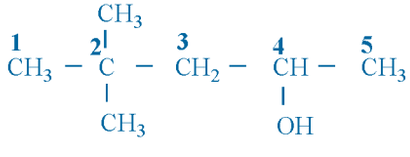


2,3-Dimethyl-5-ethyl heptane (wrong)

Rule 4: Lowest number for functional group: When the functional group is present in the chain, it should be given first preference even if it violates lowest number rule 2. Double bond or triple bond also considered as functional groups.



4,4-Dimethyl-2-pentanol (Correct)



2,2-Dimethyl-4-pentanol (Wrong)

The order of preference of numbering is as follows.

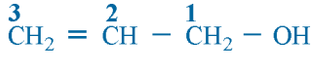
(i) To the principal functional group of a compound.
(ii) To the double or triple bond.
(iii) To the substituent atoms or groups.

When more than one functional group present in the compound, then the order of preference is as follows.

1. Carboxylic acids
2. Carboxylic acid derivatives
3. Aldehydes
4. Nitriles
5. Ketones

6. Alcohols
7. Amines
8. Ethers
9. Olefins
10. Acetylenes

Systematic name for allyl alcohol is



2-Propene-1-ol

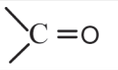
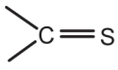
For nomenclature purpose the following functional groups are considered as substituents not as functional groups (halo, nitroso and azo as they do not have ending).

When there is more than one functional group in the compound, one is principal functional group and the other is secondary functional group. The prefixes and suffixes used for various functional groups are depicted in the following Table 2.1.

Table 2.1 Groups cited only as prefixes.

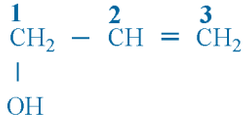
S. No.	Functional Group	*Prefix* name
1.	– F	Fluoro
2.	– Br	Bromo
3.	– Cl	Chloro
4.	– ClO	Chlorosyl
5.	– ClO_2	Chloryl
6.	– ClO_3	Perchloryl
7.	– I	Iodo
8.	=N_2	Diazo
9.	– N_2	Azido
10.	– NC	Carbylamino
11.	– NO	Nitroso
12.	– NO_2	Nitro
13.	– N(O)OH	*aci*-Nitro
14.	– OR	Alkyl or aryl-oxy
15.	– SR	Alkyl-thio
16.	$-OC\equiv N$	Cyanato
17.	– OOH	Hydroperoxy
18.	– OR	Alkyl-oxy
19.	– OOR	Alkyl-dioxy
20.	$-S-C\equiv N$	Thiocyanato
21.	– SR	Alkyl-thio
22.	– S(O)R	Alkyl-sulphinyl
23.	– SO_2R	Alkyl-sulphonyl
24.	– SSR	Alkyl-dithio
25.	$-N=C=O$	Carbonylamino
26.	$-N=C=S$	Thiocarbonylamino

Table 2.2 Groups cited as prefixes or suffixes.

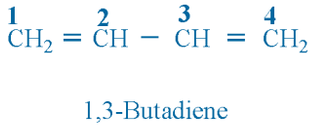
S. No.	Functional Group	*Prefix* name	*Suffix* Name
1.	—COOH	Carboxy	Carboxylic acid
2.	– SO_2 OH	Sulpho	– sulphonic acid
3.	—COX	---	– oyl(-yl) halide
4.	—$CONH_2$	Carbamoyl	– carboxamide or amide
5.	—C≡N	Cyano	– nitrile
6.	$\overset{\oplus}{-}N\equiv\overset{\ominus}{C}$	Cyano	– Isonitrile
7.	—CHO	Formyl	– al
8.		Oxo	– one
9.		Thioxo	– thione
10.	—OH	Hydroxy	– ol
11.	—SH	Mercapto	– thiol
12.	$-NH_2$	Amino	– amine
13.	=NH	Imino	– imine

Rule 5: Writing names for compounds containing more than one functional group: Whenever more than one functional group are present in the given compound then the ending is suitably modified. Carbon – carbon multiple bonds and second functional groups are combined in endings or the important functional group is considered as substituent.

Some of the examples are shown below.



2-Propene-1-ol



1,3-Butadiene

Name of some compounds containing two or more functional groups are shown in Table 2.3.

Table 2.3 Nomenclature of some simple polyfunctional compounds.

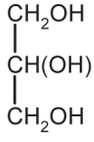
S. No.	Functional Groups	Generic name	Specific examples	
			Structure	Name
1.	Two hydroxyl	Alkanediol		Ethane-1,2-diol
2.	Three hydroxyl	Alkanetriol		1,2,3-Propane triol
3.	Two double bonds	Alkadienes	CH_2=CH—CH=CH_2	1,3-Butadiene

Table 2.3 *Contd...*

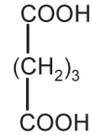
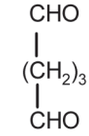
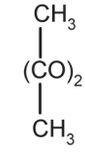
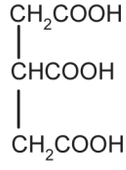
S. No.	Functional Groups	Generic name	Specific examples	
			Structure	Name
4.	Two acids	Alkanedioic acid		Pentanedioic acid
5.	Two aldehydes	Alkanedial		Pentanedial
6.	Two ketones	Alkanedione		2,3-Butanedione
7.	Three acids	Tricarboxylic acid		1,2,3-Propane tricarboxylic acid

Table 2.4 Nomenclature of some unsaturated compounds of Simple Functions.

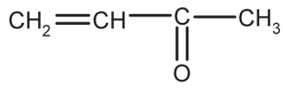
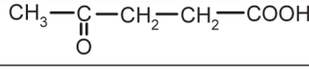
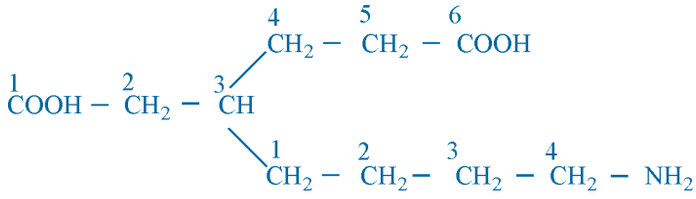
S. No.	Unsaturation and Functional Groups	Generic name	Specific examples	
			Structure	Name
1.	Double bond and acid	Alkenoic acid	$CH_2{=}CH{-}COOH$	Propenoic acid
2.	Triple bond and aldehyde	Alkynal	$CH{\equiv}C{-}CHO$	Propynal
3.	Double bond and ketone	Alkenone		3-Butene-2-one
4.	Two double bonds and alcohol	Alkadienol	$(CH_2{=}CH{-}CH_2)_2\,CHOH$	1,6-Hepatadien-4-ol
5.	Double bond and two hydroxyls	Alkenediol	$HOCH_2{-}CH{=}CH{-}CH_2OH$	2-Butene-1,4-diol

Table 2.5 Nomenclature of some compounds of complex Functions.

S .No.	Complex functional Groups	Generic name	Specific examples	
			Structure	Name
1.	Keto acid	Oxoalkanoic acid		4-Oxopentanoic acid
2.	Hydroxy acid	Hydroxy alkanoic acid	$HOCH_2{-}COOH$	Hydroxy ethanoic acid
3.	Amino acid	Amino alkanoic acid	$H_2N(CH_2)_5COOH$	6-Amino hexanoic acid
4.	Cyano acid	Cyano alkanoic acid	$N{\equiv}C{-}CH_2{-}COOH$	Cyano ethanoic acid
5.	Aminoketone	Aminoalkanone	$H_2N(CH_2)_2COCH_3$	4-Amino-2-butanone
6.	Alkoxy alcohol	Alkoxy alkanol	$CH_3O{-}CH_2{-}CH_2OH$	2-Methoxy ethanol

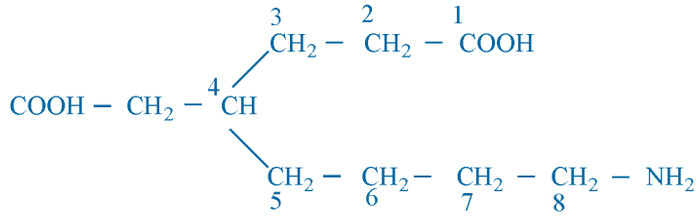
Rule 6: Treatment of "like things alike": All groups of one kind which occurs in a single molecule should be given the same treatment as far as possible.

For example, in the given example carboxylic acid is the main functional group, the parent compound should include two or three functional groups as possible.



3-(4-aminobutyl)hexane-1,6-dioic acid

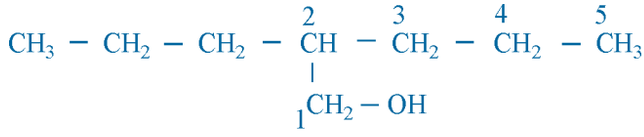
(Preferred)



8-Amino-4-(carboxyl methyl) octanoic acid

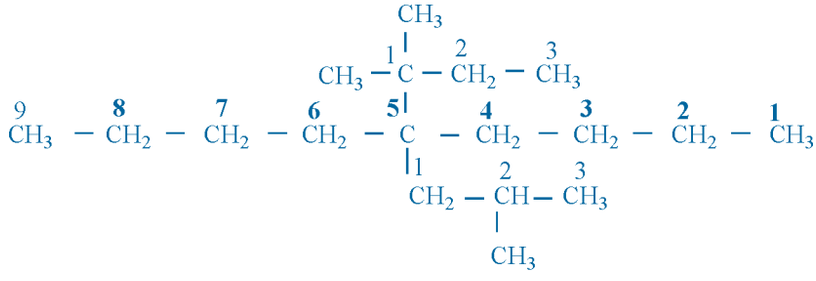
(Not preferred)

Rule 7: Functional groups and the selected chain: Maximum number of functional groups must be included in the carbon chain even if it violates longest chain rule (Rule 1), as shown in the following example.



2-Propyl-1-pentanol

When there is a side chain with side chain, the latter is numbered and the name of the complex is considered to start with the first letter of its complete name, as shown in the following example.



5-(1,1- Dimethylpropyl)-5-(2-methylpropyl) nonane

In addition to these rules, following points mentioned are also useful in writing IUPAC name of compound.

Steps involved in writing IUPAC name of the compound

Step 1: Locate the longest chain containing principal functional group and as many as secondary functional group and carbon-carbon multiple bonds.

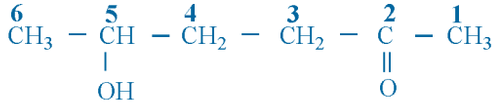
Step 2: Select the root word corresponding to the chain length. For example Hex for six carbon atom chain.

Step 3: Number the longest chain selected from the end near to the principal functional group.

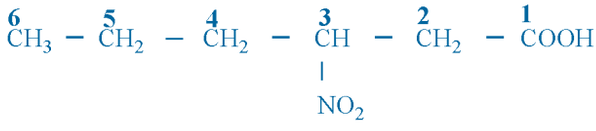
Step 4: Based on the carbon-carbon bonds $C-C$, $C=C$, $C\equiv C$ attach the suffix -ane, -ene or yne respectively to the root word of carbon chain.

Step 5: Add suitable prefixes and suffixes with numerals to indicate the number and position of each side chain, substituent, or functional group.

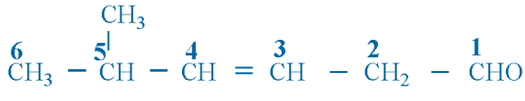
Example:



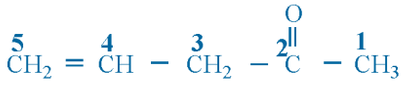
5-Hydroxy-2-hexanone



3-Nitrohexanoic acid



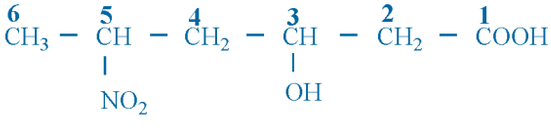
5-Methyl-3-hexen-1-al



4-Pentene-2-one

(or)

Pent-4-en-2-one



3-Hydroxy-5-nitro hexanoic acid

Notes:

1. **Position of numerals used in the enumeration of substituents:** Numerals representing location of unsaturation or functional groups are placed before the name stem as

2-Pentene	not pentene-2
1-Chloro-2-pentene	not 1-chloropentene-2
1-hexene-3-yne	Not hexenyne-3.

2. **Writing names:** The names of radical replacing hydrogen atom in compound are carried out. For example, Chlorotoluene (chlorine replaced "H" atom of toluene).
   Elision of vowels: To avoid ambiguity vowels, whether pronounced or silent are generally retained in systematic naming. This results in using of double vowels, e,g, cyclooctane.

However it has been accepted to elide following vowel a,e and o in the following circumstances.

(i) When preceding the suffix name of a functional group, example
Propanol not propaneol
Hexamine not hexane amine.

(ii) Naming Alkenes and Alkynes on the same compound-example Pentenyne not penteneyne.

3. **Punctuation marks:** Most commonly used punctuation in naming organic compounds are hyphens, commas and enclosing brackets.

(i) ***Hyphens:***

(a) Used to connect numbers and letters serving as a locants.
For example: 2-Chloropropanone 1-Bromo-3-chlorobutane.

(b) Used to connect the prefixes like *cis, trans* (configurational prefixes) or structural prefixes (*sec, tert, neo*) with the compound name.
For example: *Cis*-2-butene, tert–butyl alcohol
The prefixes *cis, trans, iso, neo, tert etc* should be in italics.

(ii) ***Commas:*** Used to separate individual members of a series of locants.
Example: 1,1,2-Trichloro propane.

(iii) ***Enclosing brackets:*** Parenthesis ( ) and square [ ] are used as demarcation symbols when the locants are related to complete names. Example: 4-amino-N-(hydroxyl ethyl) butyramide.

Writing the structural formula from the given IUPAC name: To write the chemical strcture of a given compound from the IUPAC name, the following steps are to be adopted.

(i) ***Locate the parent alkane:*** From the name write the number of carbon atoms of the alkane in a straight chain and number them from any one of the end.

(ii) ***Locate the suffix:*** Locating suffix gives information about chain length, nature of functional group along with the positions.

(iii) ***Locate the groups / substituents:*** As mentioned in prefix locates the groups position in the chain.

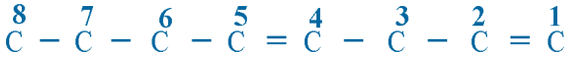
(iv) ***Add hydrogen atoms if required to satisfy:*** Four valencies of each carbon atoms to get the formula.

Thus, for writing the structural formulae of 3-ethyl-2,5-dimethyl-1,4-octadiene.

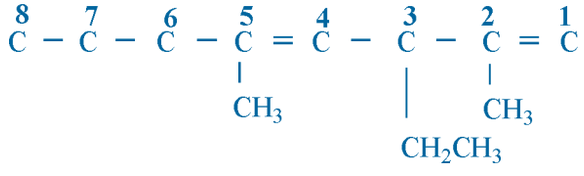
(i) Parent alkane is octane. Write eight carbon atoms in a straight chain and number it.



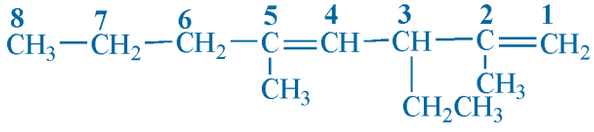
(ii) The suffix diene indicates two double bonds in 1 and 4 points.



(iii) To locate the groups mentioned in prefix we attach ethyl group on 3rd C and methyl groups to 2nd C and 5th C to get the desired compounds.



(iv) Finally to satisfy valencies hydrogen atoms are added.



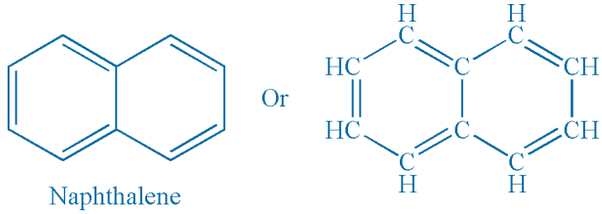
3-Ethyl-2,5-dimethyl-1,4-octadiene

25

Polynuclear Hydrocarbons

Introduction

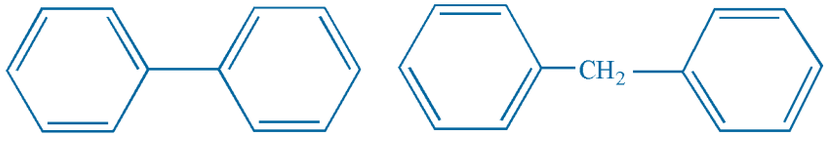
Polynuclear hydrocarbons or condensed nuclear hydrocarbons are compounds in which two or more carbon atoms are shared commonly by two or more aromatic rings. Simple examples of these type of compounds are naphthalene, anthracene, phenanthrene and their derivatives. In naphthalene two carbon atoms are shared by two benzene rings as shown below.



Naphthalene Naphthalene

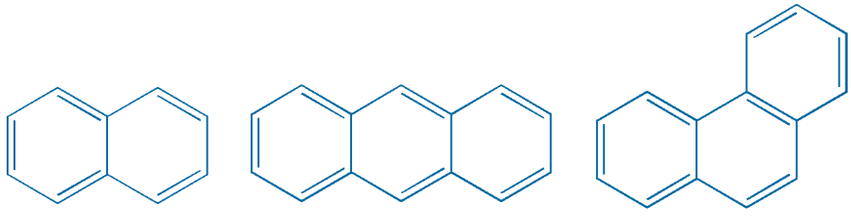
Types: Polynuclear hydrocarbons are classified into two types.

1. **Compounds in which the rings are isolated,** example: Biphenyl, diphenyl methane *etc.*



Biphenyl Diphenyl methane

2. **Compounds in which the two or more rings are fused in *o*-positions:** Naphthalene, anthracene and phenanthrene.



Naphthalene Anthracene Phenanthrene

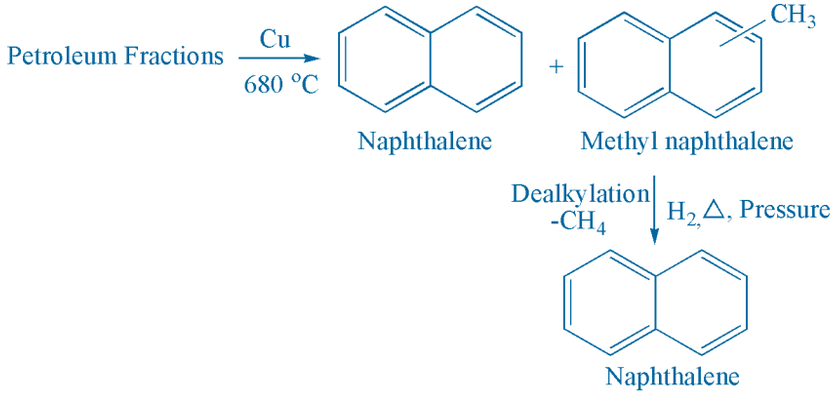
Naphthalene

Manufacture of naphthalene: Major source of naphthalene is coal tar. Coal tar contains 6-10% of naphthalene and it is present in the middle oil fraction of coal tar and obtained by distillation process. Allow the middle oil to cool, maximum amount of naphthalene is crystallized out and is collected by

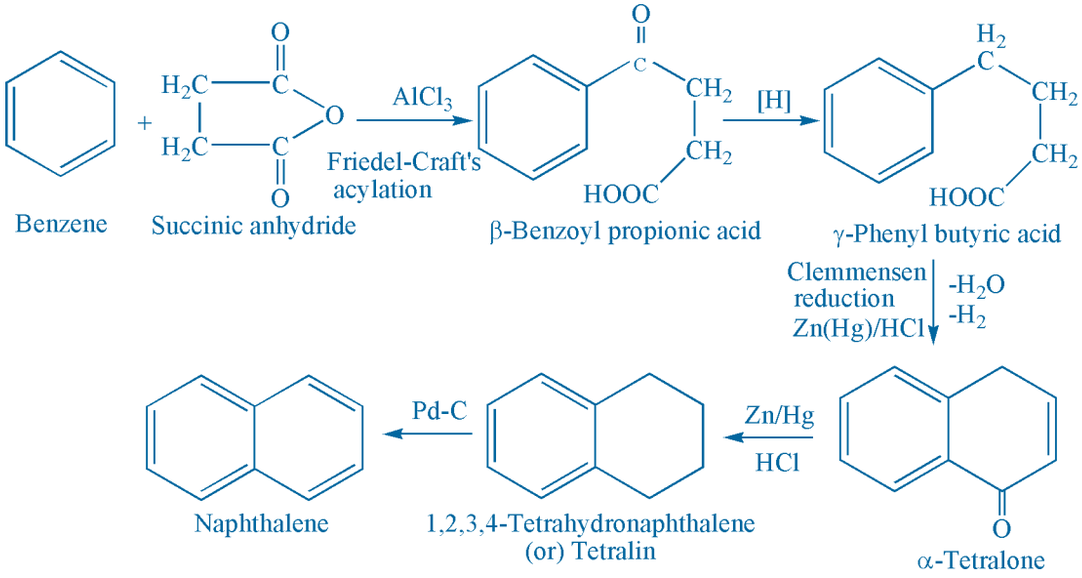
centrifugation or pressing out the oil in a hydraulic press. The crystals obtained are washed with water and with sodium hydroxide solution in a centrifuger to remove adhered oil and phenols. Then it is treated with conc. H_2SO_4 to remove the alkaline impurities, the crude naphthalene obtained is purified by sublimation and further purified by recrystallisation with petroleum ether. Nowadays the "hot processing process" is replaced by continuous washing or distillation.

Synthesis

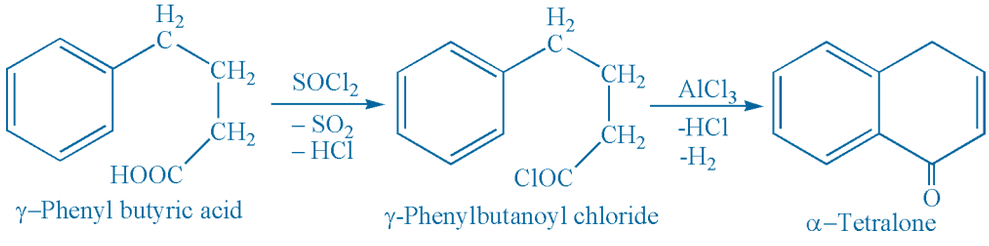
1. **From petroleum fractions:** Petroleum fractions are passed over a heated copper catalyst at 680 °C at atmospheric pressure to give naphthalene and methyl naphthalene and the later undergoes hydro dealkylation to give naphthalene.



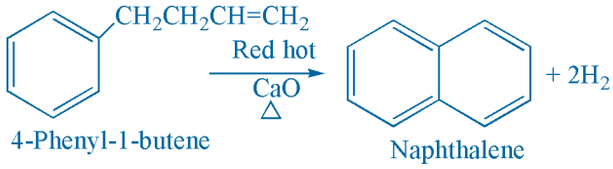
2. **Haworth synthesis:** Benzene is treated with succinic anhydride followed by reduction to yield γ-phenyl butyric acid. The later one undergoes ring closure reaction in the presence of conc. H_2SO_4 to yield α-tetralone. Reduction of α-tetralone with Zn(Hg)/HCl yields tetrahydronaphthalene (tetralin) which upon further dehydrogenation yields naphthalene by heating with selenium or palladised charcoal. The sequence of the chemical reactions are as follows.



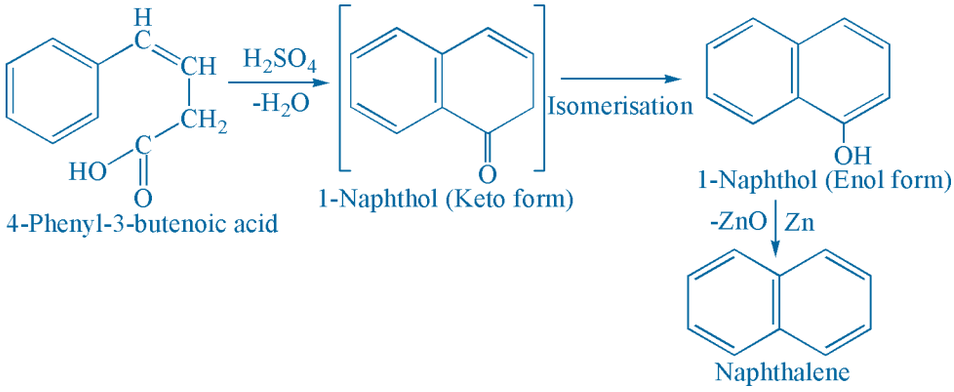
Ring closure is also effected by Friedel-Crafts reaction on acid chlorides as shown below.



3. **From 4-phenyl-1-butene:** When 4-phenyl-1-butene is passed over red-hot calcium oxide naphthalene is obtained.



4. **From 4-phenyl-3-butenoic acid:** When 4-phenyl-3-butenoic acid is warmed with conc. H_2SO_4 1-napthol is formed which upon further distillation with Zn dust yields naphthalene.



Physical Properties: It is a colorless crystalline substance with characteristic odour (moth ball odour). It is very volatile and readily sublimes on heating. It is insoluble in water but soluble in organic solvents.

Chemical Properties: Chemical properties of naphthalene resembles to that benzene. It is less aromatic than benzene and forms substitution products more readily than benzene. Like alkenes it forms addition products readily than benzene. But as soon as one of the rings is saturated or destroyed by oxidation, the second ring is stable as the benzene ring. The important characteristic reactions of naphthalene are as follows.

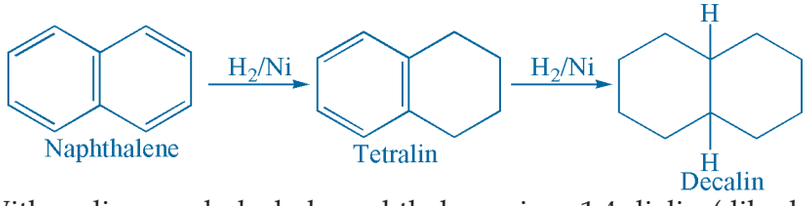
(I) Addition reactions.

(II) Electrophilic aromatic substitution reactions.

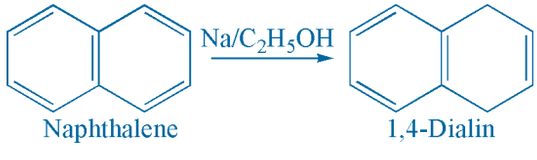
(I) Addition reactions

1. **Addition of hydrogens:** Naphthalene gives different kind of products depending upon the type of reducing agents used.

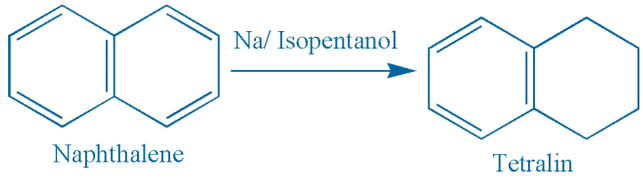
(a) With catalytic reduction using nickel yields decalin or decahydro naphthalene



(b) With sodium and alcohol, naphthalene gives 1,4-dialin (dihydronaphthalene).

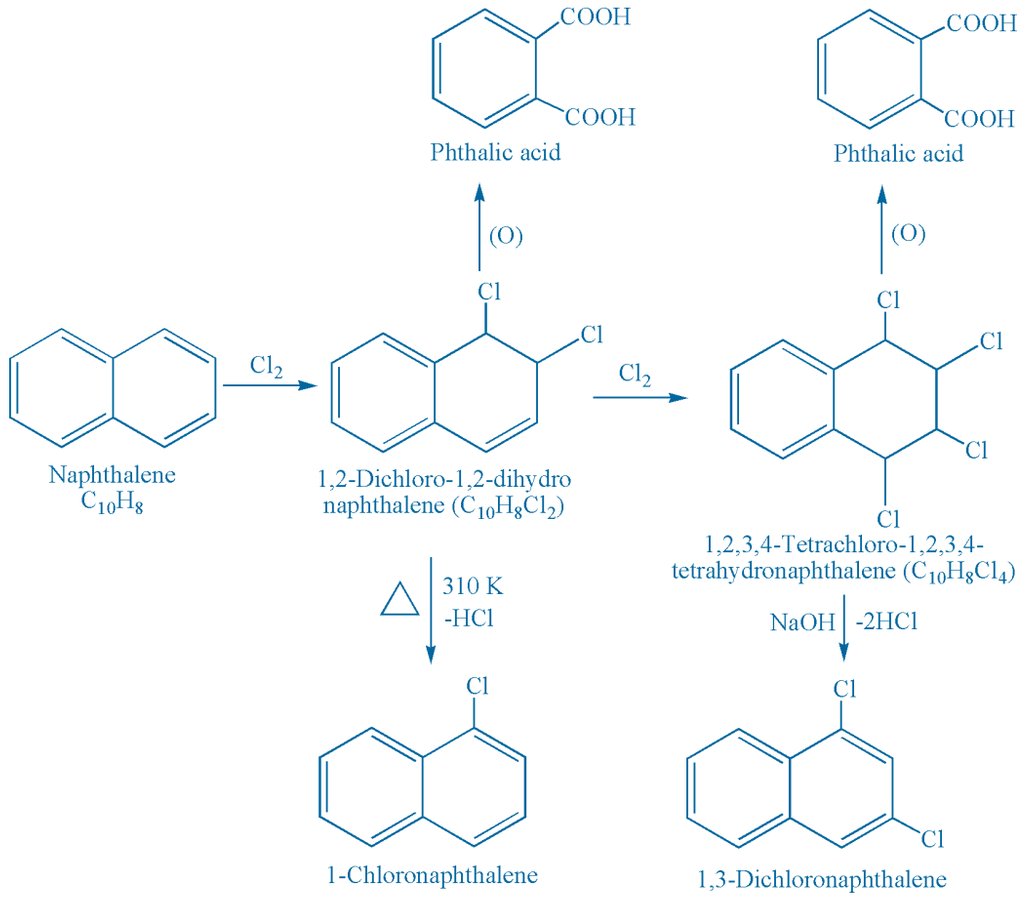


(c) With sodium and isopentanol naphthalene gives 1,2,3,4- tetrahydro naphthalene or tetralin.

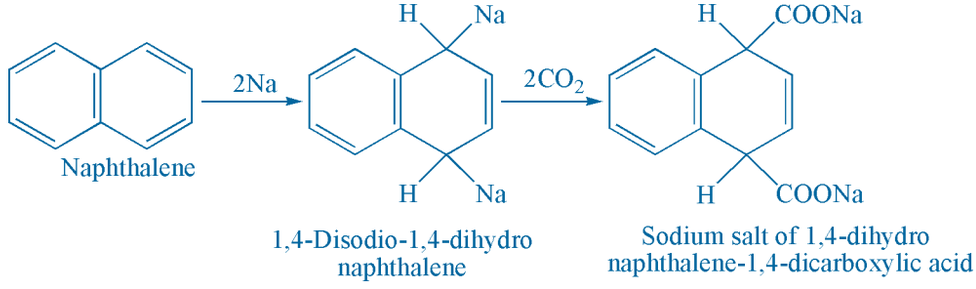


2. **Addition of chlorine:** Solid naphthalene reacts with dry chlorine to give naphthalene di and tetra chlorides. Both of them undergo oxidation to yield phthalic acid. It indicates that the halogen atoms are present in the same ring.

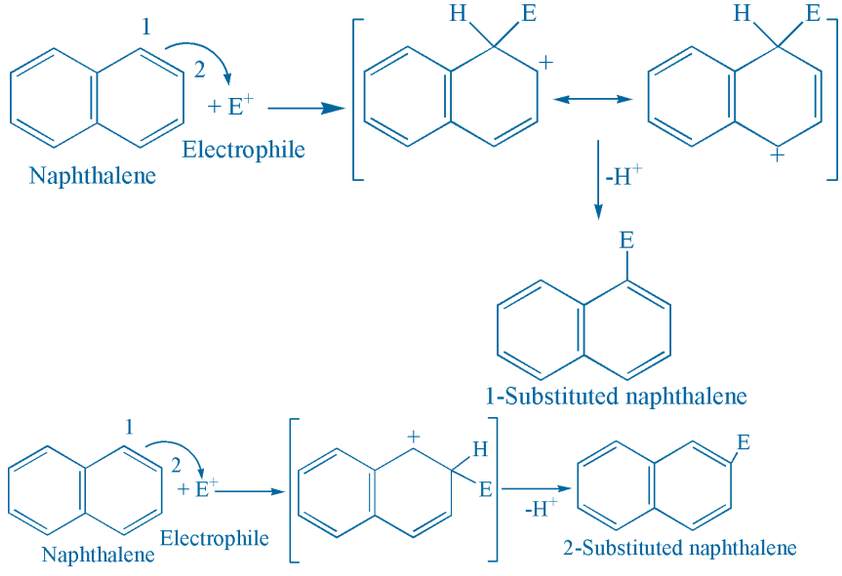
The naphthalene dichloride when heated at 310 K, loses one molecule of hydrogen chloride and gives 1-chloro naphthalene. Naphthalene tetrachloride on treatment with alkali gives dichloro naphthalene.



3. **Addition of sodium:** Naphthalene upon reaction with sodium gives 1,4-disodio naphthalene which reacts further with carbon dioxide with the formation of sodium salt 1,4-dihydro naphthalene-1,4-di carboxylic acid.

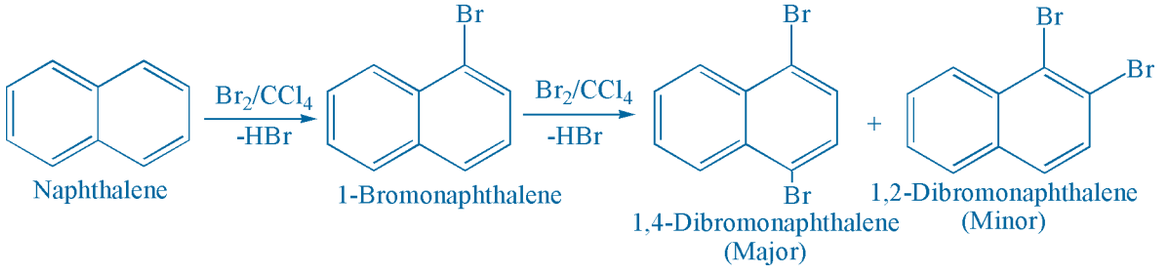


(II) Electrophilic aromatic substitution reactions: Naphthalene undergoes electrophilic aromatic substitution reactions and the major substitution occurs at C_1 (α-position) and C_2 positions.

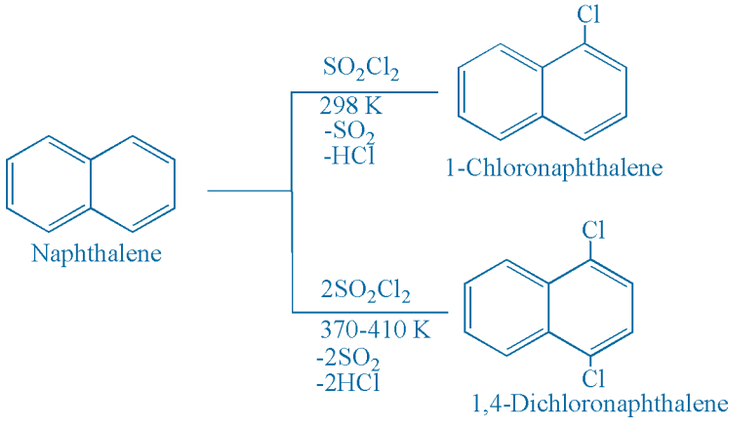


The product by C-1 attack is always predominates because the carbonation intermediate obtained by C-1 attack is more stable (due to resonance), the C-2 attack is possible only when the reaction occurs at high temperature or when bulkier solvents are used.

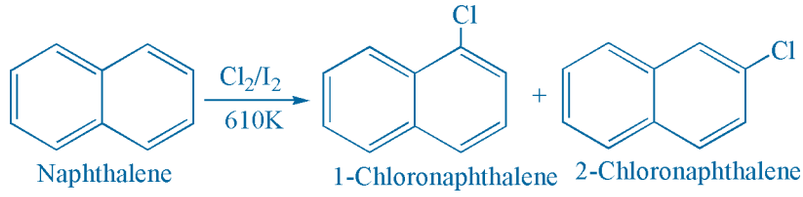
1. **Halogenation (Bromination):** Naphthalene reacts with bromine in boiling carbon tetrachloride solution to give 1-bromonapthalene. Further bromination gives mainly the 1,4-dibromo naphthalene with small amount of 1,2-dibromo naphthalene.



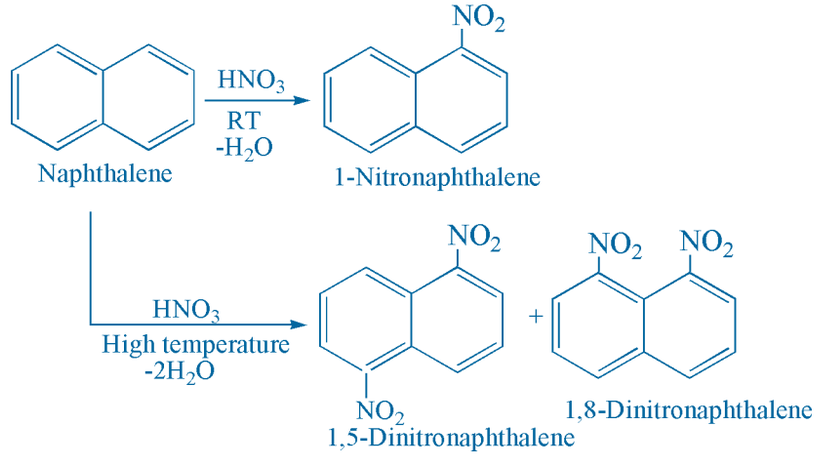
Chlorination: Naphthalene reacts with sulphuryl chloride in the presence of aluminium chloride. Naphthalene reacts with one equivalent of sulphuryl chloride at 298 K to give 1-chloro naphthalene and with two equivalents of sulphuryl chloride at 370-410 K to gives 1,4-dichloronaphthalene.



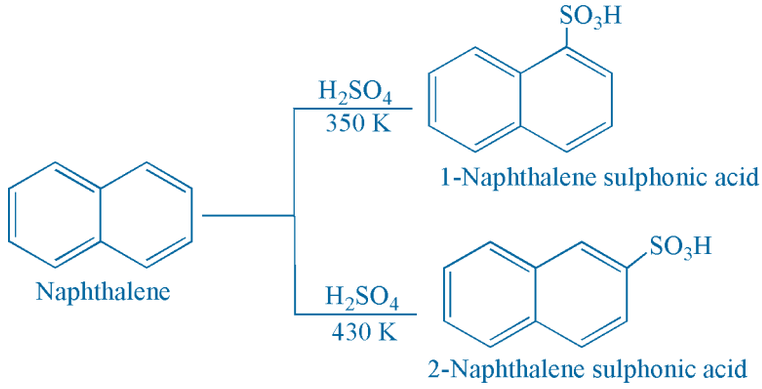
Naphthalene reacts with chlorine in the presence of iodine above 610 K and gives a mixture of 1 and 2-chloro naphthalene.



2. **Nitration:** Naphthalene reacts with nitric acid at room temperature to yield 1-nitro naphthalene and at high temperature yield 1,5 & 1,8-dinitro naphthalene.

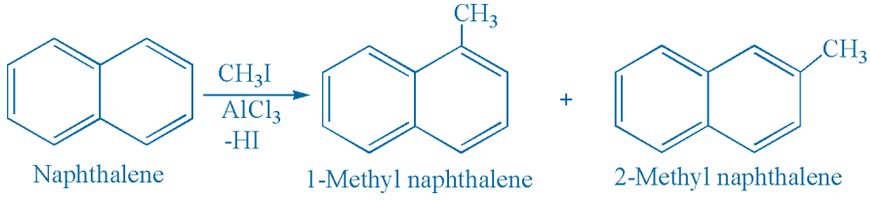


3. **Sulphonation:** Naphthalene reacts with conc. H_2SO_4 at 350 K and yields 1-naphthalene sulphonic acid, but at 430 K it gives 2-naphthalene sulphonic acid.

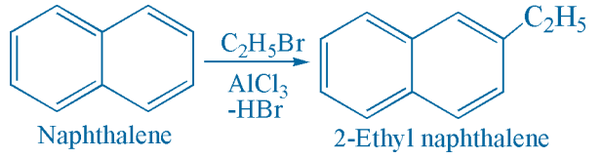


4. **Fridel-Craft's reaction:** Carry out this reaction at low temperature in the presence of anhydrous aluminium chloride. Because at high temperature, one of the naphthalene ring opened.

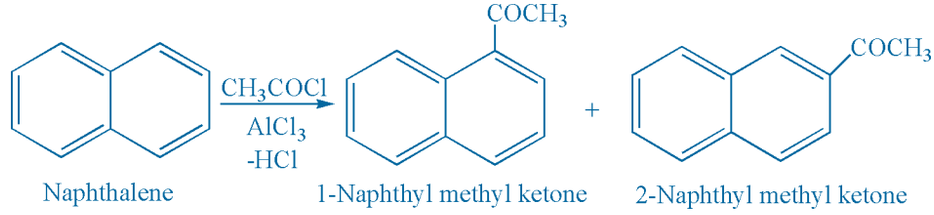
   (a) With methyl iodide: Naphthalene reacts with methyl iodide in the presence of anhydrous $AlCl_3$ and gives 1 & 2-methyl naphthalene.



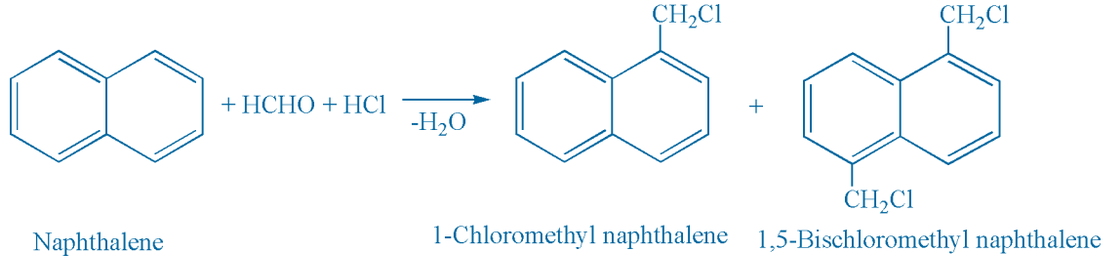
But naphthalene reacts with ethyl bromide to yield 2-ethyl naphthalene.



(b) With acetyl chloride: Naphthalene reacts with acetyl chloride and $AlCl_3$ to yield a mixture of 1-naphthyl methyl ketone & 2-napthyl methyl ketone. The composition of mixture obtained depends upon the nature of the solvent and the temperature used. For example, in the presence of CS_2 at 260 K, the 1 and 2 - derivatives are obtained in 3:1 ratio while in the presence of nitrobenzene at 298 K the product obtained is in 1:9 ratios.



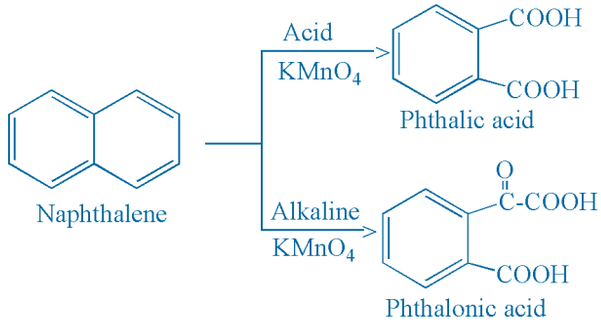
5. Chloro methylation: Naphthalene reacts with formaldehyde/HCl and glacial acetic acid, gives 1-chloro methyl naphthalene with a small amount of 1,5-bis chloromethyl naphthalene.



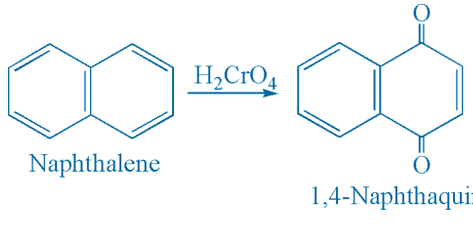
6. Oxidation: Naphthalene gives different kind of products on oxidation. The products formed depend upon the nature of the oxidizing agents used.

The various kind of oxidation reactions are as follows.

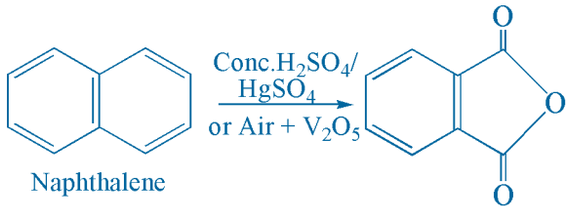
(i) With potassium permanganate: Naphthalene reacts with acidic $KMnO_4$ and alkaline $KMnO_4$ to give phthalic acid and phthalonic acid respectively.



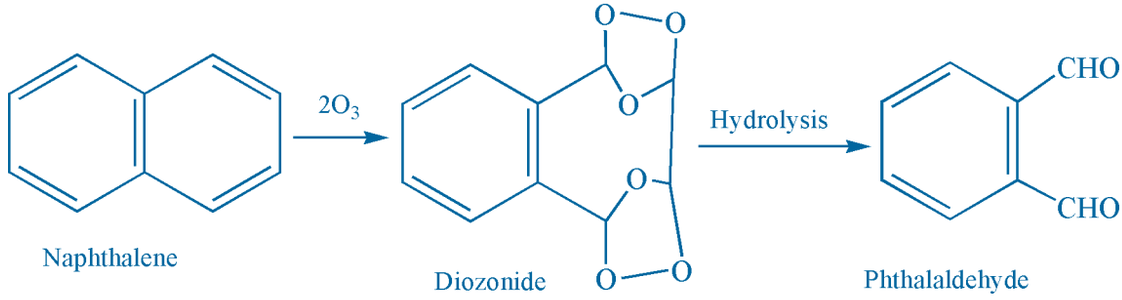
(ii) With chromic acid: Naphthalene when oxidised with chromic acid yields 1,4-naphthaquinone.



(iii) Naphthalene reacts with conc. H_2SO_4 and mercuric sulphate or air in the presence of vanadium pentoxide and yields phthalic anhydride.



(iv) With ozone: Naphthalene is oxidized with ozone to give diozonide which upon hydrolysis gives phthalaldehyde.



Uses:

1. It is used as an insecticide and for preventing moths in clothes.
2. Large amount of naphthalene is used in industry for the manufacture of various dye stuffs such as indigo, azo dye and eosin.
3. It is also used for manufacturing of phthalic anhydride, phthalic acid and phthalimide *etc.*
4. Local gas is carbureting with naphthalene.

Medicinally useful compounds containing naphthalene

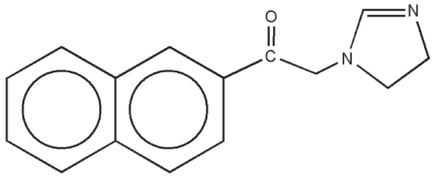
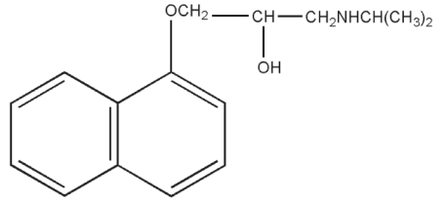
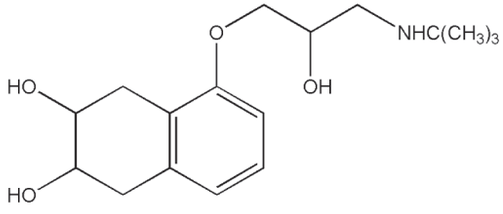
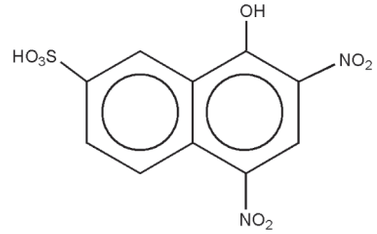
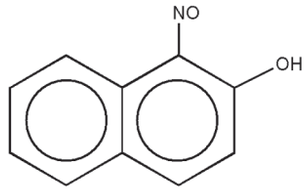
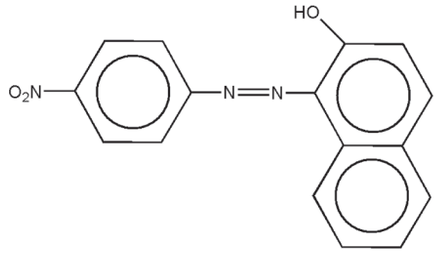
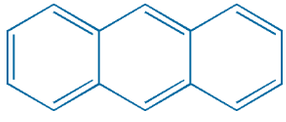
Nafimidone  **Use:** It is used as an anticonvulsant.	**Propranolol**  **Use:** It is used in the treatment of tachycardia.
Nadolol  **Use:** It is used as an antihypertensive drug.	**Naphthol yellow S**  **Use:** It is used as a dye.

Table *Contd...*

Mordant green 4	Para red azo dye
	
Use: It is used as a dye.	**Use:** It is used as a dye.

Anthracene

Anthracene is another example of fused aromatic hydrocarbons in which three benzene rings fused together in the *o*-positions.



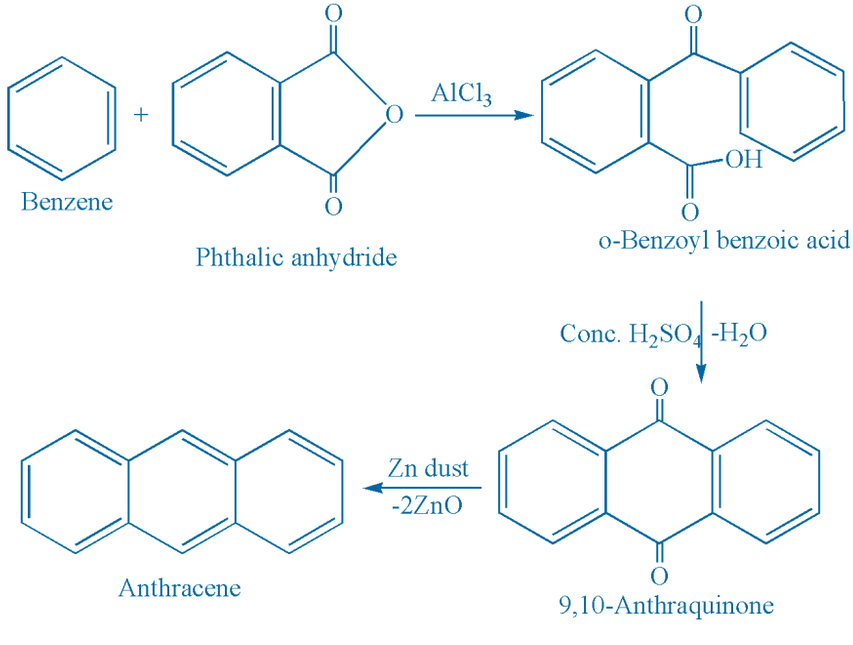
Anthracene is present in coal tar less than 1%. The name derives from the Greek word anthraz (means coal). High boiling point fraction of coal tar also contains anthracene, hence called as anthracene oil.

Preparation of anthracene: Coal tar is a chief source of anthracene and it contains 0.25-0.45% of anthracene. It is present in anthracene oil or green oil along with phenanthrene, carbazole and other substances.

Anthracene oil is allowed to stand in shallow tanks where by a viscous mass separates out and the crude anthracene (20%) is removed by filtration. During this, the anthracene content increases up to 35% and some amount of oil is removed by centrifuge when 50% anthracene is obtained. The resulting product is powdered and washed with solvent naphtha which dissolves out phenanthrene and then treated with pyridine which removes carbazole. Anthracene is finally crystallized out from benzene.

Synthesis

1. **Haworth synthesis:** Benzene reacts with phthalic anhydride in the presence of $AlCl_3$ to yield o-benzoyl benzoic acid, which upon reaction with conc. H_2SO_4 to yield 9,10-anthraquinone. The later obtained is distilled with Zn, and yields anthracene.

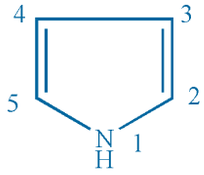


29

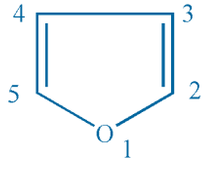
Heterocyclic Compounds (Part 1)

Introduction to Heterocyclic compounds

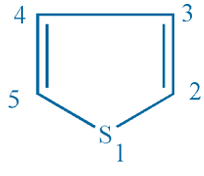
Heterocyclic compounds are cyclic compounds in which the ring carbon atoms are displaced or substituted with one or more polyvalent atoms. The major substituted polyvalent atoms are nitrogen, oxygen and sulphur as shown in the following examples.



Pyrrole



Furan

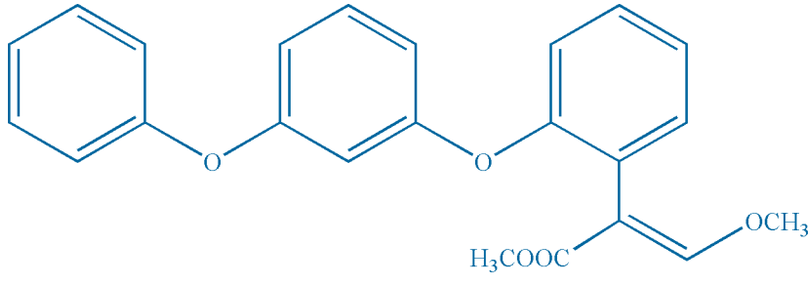


Thiophene

Heterocycles possess wide range of applications in day to day life, such as drugs, veterinary products and agrochemicals. They are also used as antioxidants, dyestuffs, pigments and additives *etc*.

Why heterocyclic compounds are very important in organic chemistry?

1. "Easy manipulation of ring system is possible to attain a required modification leads to the formation of different structural variations with different properties. This phenomenon is an useful strategy for developing new drugs. For example, consider the following compound I, which acts as a fungicide and it is highly lipophilic in nature; the water solubility of the compound is increased by the replacement of the benzene ring with suitable heterocycles.



(I)

2. Another important fact is we can easily accommodate any functional groups into the heterocycles framework as a substituent or as a part of the ring system itself. It makes heterocyclic chemistry as a very peculiar part in organic chemistry.
3. Heterocycles or heterocyclic compounds are widely distributed in nature as important fundamental units of living systems as shown below with examples.
   (i) Nucleic acid contains pyrimidine and purine rings, which are responsible for cell replication.

(ii) Chlorophyll and heme contains porphyrin ring system which is needed for photosynthesis and for respiration (oxygen transport) in higher plants and animals respectively.

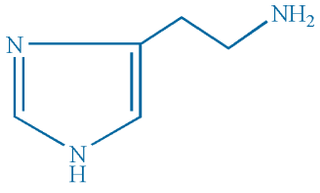
(iii) Vitamins: Thiamine (vitamin B_1), riboflavin (vitamin B_2), pyridoxol (vitamin B_6), ascorbic acid (vitamin C) - All are heterocyclic compounds.

(iv) Amino acids: Histidine, prolines, and tryptophan are possesing heterocyclic rings.

Pharmaceutical applications of heterocyclic compounds: Many of the drugs are heterocyclic in nature and possess heterocyclic ring skeleton. They are not extracted from the nature due to the difficulties in extraction and purification process. Hence they are manufactured or synthesized in the laboratories.

The origin of organic chemistry is based on the natural products and many of the drug candidates are developed subsequently. Some of the examples are described below.

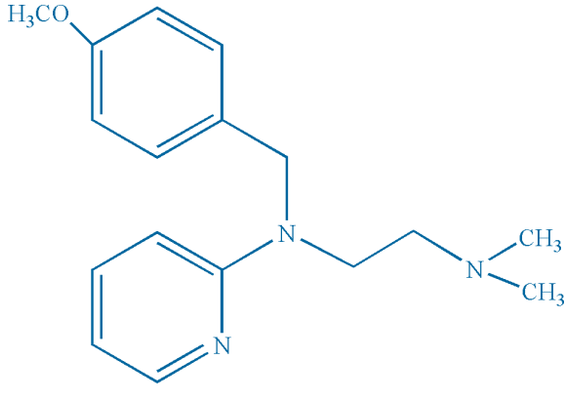
Development of Histamines and Antihistamines



(1)

Histamine

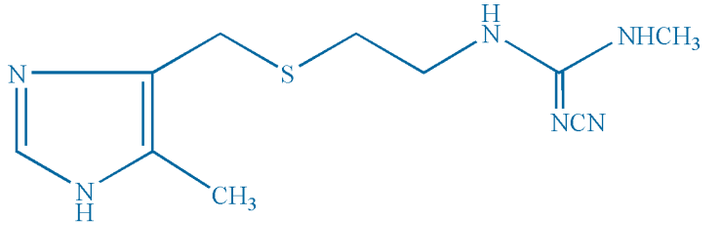
Histamine (1), a monosubstituted imidazole ring system released in our body from the amino acid histidine by decarboxylation. The main pharmacological actions of histamine includes contraction of smooth muscle, fall in BP (hypotension), producing allergic reactions and regulation of gastric acid secretion. For antagonising the action of histamine several kind of drugs are synthesized from 1940. One of the important drug is pyrilamine, a pyridine derivative (2).



Pyrilamine (2)

Pyrilamine antagonizes or inhibits the several actions of histamine but it does not block the gastric acid secretion.

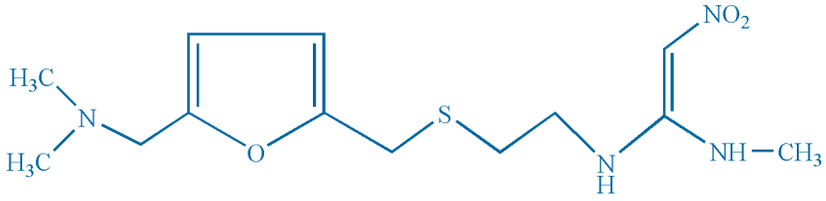
In 1976, the lacunae of pyrilamine is encountered by chemists with modification of the structure of histamine and discovered an active drug known as cimetidine (3), used for the treatment of peptic ulcer.



Cimetidine (3)

Cimetidine is one of the major leading drug in 1970s and 1980s being the first non-surgical treatment of peptic ulcer. Further development of cimetidine analogues are promoted by changing the heteroyclic ring in the cimetidine molecule.

Ranitidine, pyrrole ring of the cimetidine is replaced by furan, is also a successful drug for the treatment of peptic ulcer.

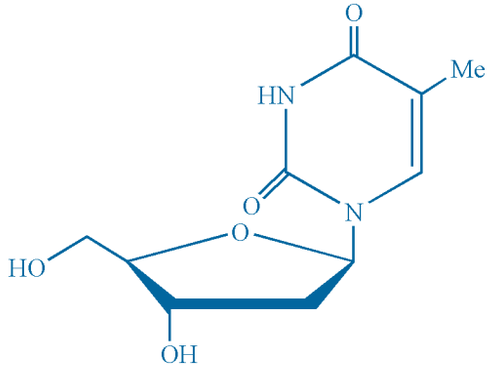


Ranitidine

Development of Nucleoside Analogues

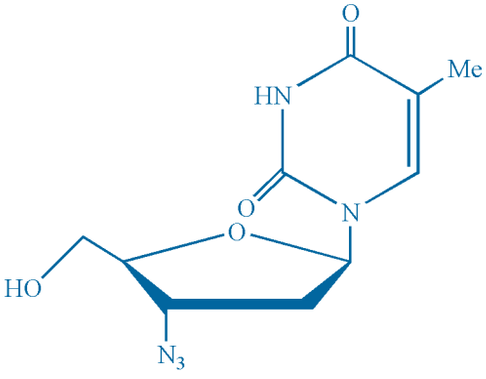
In the search for drugs to combat cancer and virus modification in the structure of DNA is the best approach by synthesizing the nucleosides.

These analogues consists of pyrimidine and purine nucleus which is attached to a sugar moiety as shown in the following examples.



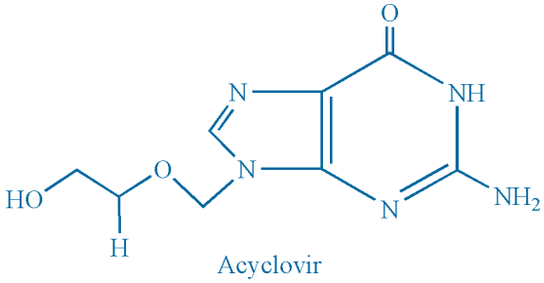
2-Deoxythymidine

Various nucleic acid analogues are developed by the structural modifications of heterocycles or sugar or both and yielded the important drugs as mentioned below.



Zidovudine

(Used in the treatment of AIDS,
Close analogue of 2'-deoxythymidine)

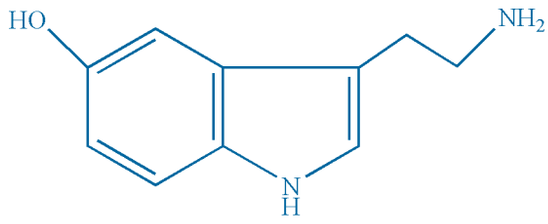


Acyclovir
(Used for the treatment of herbs virus infections, Analogue of 2'-deoxyguanosine)

Development of Alkaloidal Drugs

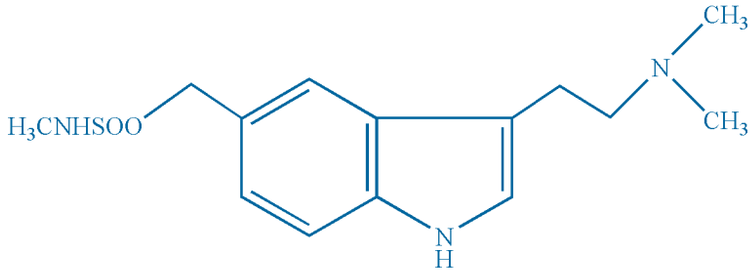
Development of Serotonin Analogues

Serotonin: Vasoconstrictor drug obtained from natural source is widely distributed in nature but present only in very low concentration. It was first extracted from natural source in 1948.



Serotonin

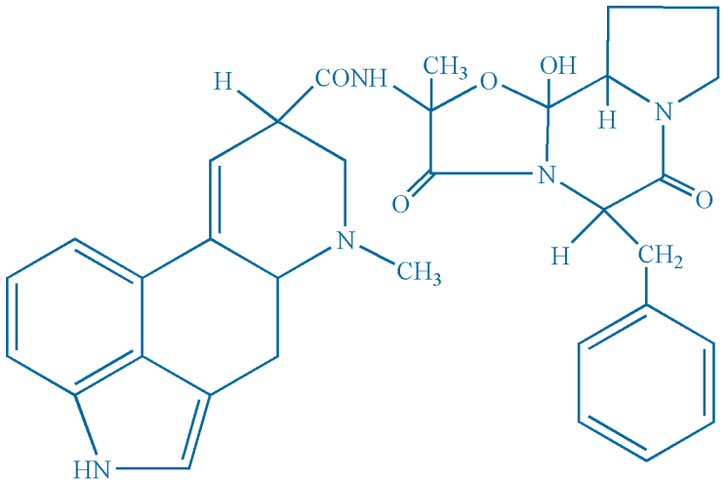
Synthesis of this drug was done in laboratories after few years and it is used for investigating the mechanism of action. The main pharmacological actions of serotonin includes constriction of brain arteries, behavourial changes in the body *etc,*. But the main drawback of serotonin is that it is too rapidly metabolised in the brain. This was overcome by designing the serotonin analogues and the simple indole derivative of serotonin such as sumatriptan was developed by researchers which acts as a selective agonist at serotonin receptor sites in the brain and used as a drug of choice for the treatment of migraine.



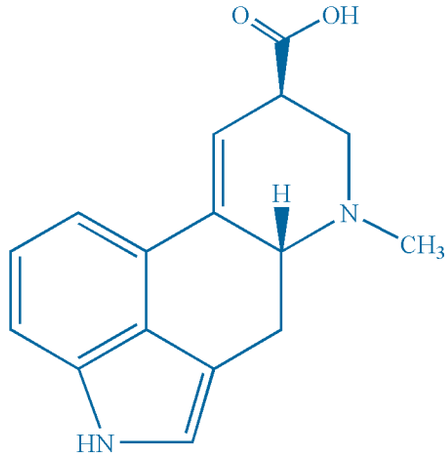
Sumatriptan
(Indole derivative of serotonin used for migraine)

Ergot alkaloids:

Ergotamine: An indole alkaloid possesing aminoethyl side chain at 3rd position, is used in migraine at low doses. But the drug is highly toxic and not in use. But its synthetic analogue lysergic acid diethylamide (LCD) is discovered and it is now notorious as a hallucinogen.



Ergotamine



Lysergic acid

Nomenclature of Heterocyclic Compounds

Heterocyclic compounds are named by using trivial names as well as systematic names. Trivial name does not provide the detailed information about the structure but still it is used (In recent years the IUPAC has made efforts to systematize the nomenclature of heterocyclic compounds).

According to the IUPAC system, following guidelines are used for naming the heterocycles.

1. The monocyclic compounds are named by a prefix which is derived from the nature of the hetero atom present in it (Eliding "a" where necessary). Some of the rings and their prefixes are indicated in the following Table 29.1.

Table 29.1 Prefix for Hetero Atoms.

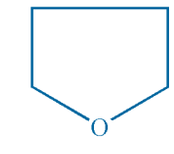
Nature of hetero atom	Symbol	Respective prefix
Oxygen	O	Oxa
Nitrogen	N	Aza
Sulphur	S	Thia
Phosphorous	P	Phospha
Selenium	Se	Selena
Silicon	Si	Sila
Germanium	Ge	Germa

2. If the same hetero atom is present more than one time, the prefixes di, tri *etc* are used.
   For example: dioxa, triaza *etc*.
3. If different heteroatoms are present in the ring, the naming starts from the atom which is high group in periodic table and as low in atomic number in that group. Hence the order of naming is as follows O, S, N, P, Si *etc*.
4. When the size of the monocyclic ring is from 3 to 10, they are indicated with suffixes which are mentioned in the Table 29.2.

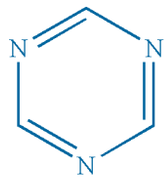
Table 29.2 Common name endings for heterocycles.

Nature of the ring size	Suffixes for completely unsaturated compounds		Suffixes for completely saturated compounds	
	With N	Without N	With N	Without N
3	-irine	-irene	-iridine	-irane
4	-ete	-ete	-etidine	-etan
5	-ole	-ole	-olidine	-olane
6	-ine	-in	-	-ane
7	-epine	-epin	-	-epane
8	-ocine	-	-ocin	-ocane
9	-onine	-onin	-	-onan
10	-ecine	-ecin	-	-ecan

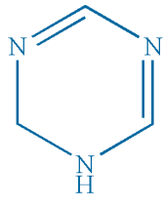
5. The nature of hydrogenation is indicated by the suffixes as mentioned in the above table or prefixes dihydro, tetrahydro *etc*. or by prefixing the parent unsaturated compound with the symbol H preceded by a number indicating the position of saturation. For example:



Tetrahydrofuran

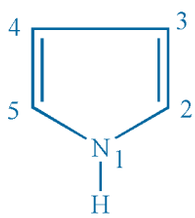


1,3,5-Triazine

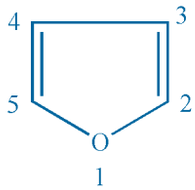


1,2-Dihydro-1,3,5-triazine

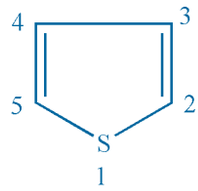
6. In monocyclic ring system with one hetero atom, numbering start from that atom only.



Pyrrole



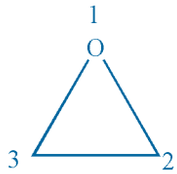
Furan



Thiophene

According to the above system, some of the heterocycles and their nomenclature are mentioned below.

Example 1:



Oxirane

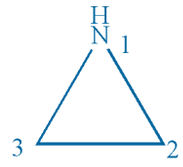
[Three membered ring without N and full saturation]



Oxirene

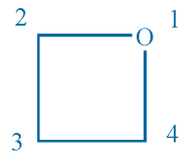
[Three membered ring without N and unsaturation]

Example 2:



Aziridine

[Three membered ring with N and saturation]

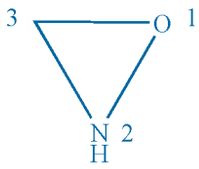


Oxetan

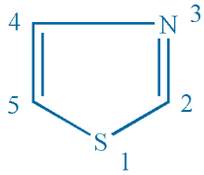
[Four membered ring with O and saturation]

Example 3:

If two or more different atoms are present in the ring, naming is given by combining the prefixes of the heteroatoms.

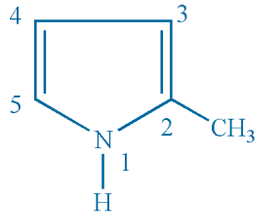


Oxa aziridine

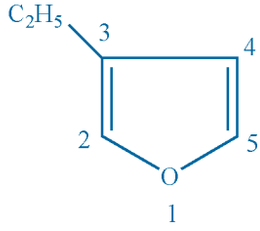


Thiazole

Nature of the substituents: In substituted heterocycles, the numbering starts from the hetero atom (assigned position 1) and the substituents are numbered with a lowest possible numbers. The name of substituents should be mentioned in alphabetical order.

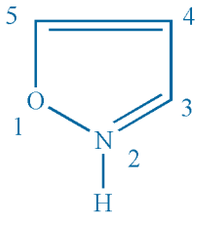


2-Methyl pyrrole



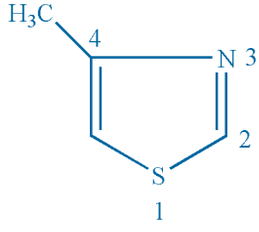
3-Ethyl furan

If the heterocycles contain more than one hetero atom, the order of preference is O, S, N. The ring is numbered from the atom of preference and preceeded in such a way so as to give the smallest possible number to the other hetero atoms in the ring. In this case, the substituents numbering need not be bothered.



[The preference of number starts from O and the next heteroatom attains lowest possible number]

Isoxazole

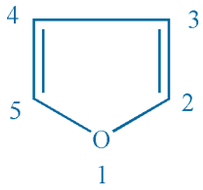


[The preference of numbering of heteroatom starts from S and the next heteroatom attains lowest possible number, no preference for subtituents]

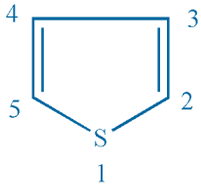
4-Methyl thiazole

7. Apart from the systematic method, many number of heterocycles are also named by common names or non-systematic names which are widely used. Few examples are shown below.

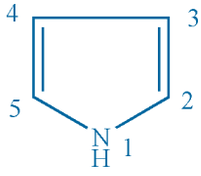
Five membered rings:



Furan

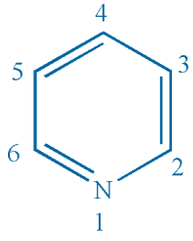


Thiophene

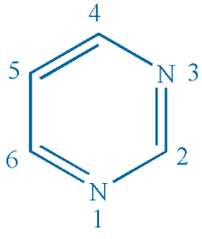


Pyrrole

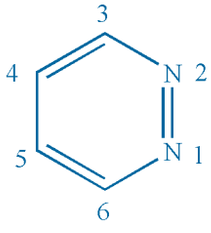
Six membered rings:



Pyridine

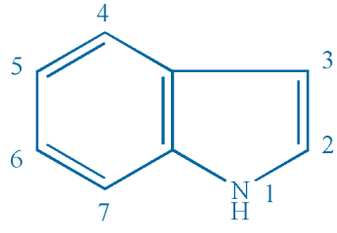


Pyrimidine

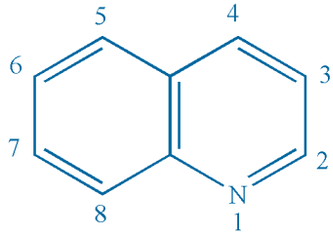


Pyridazine

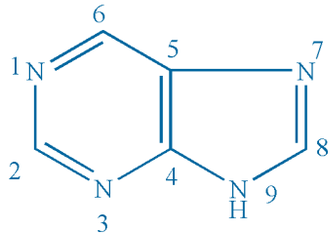
Condensed heterocycles:



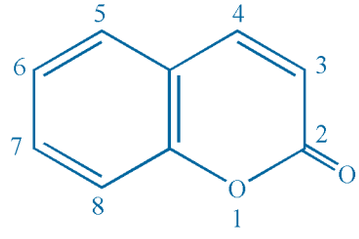
Indole



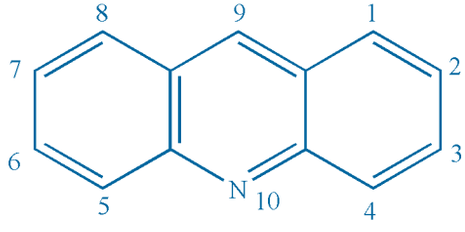
Quinoline



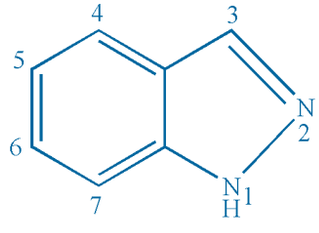
Purine



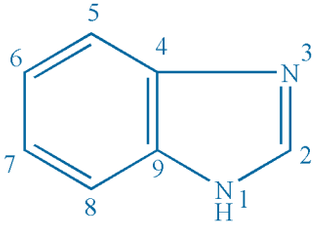
Coumarin



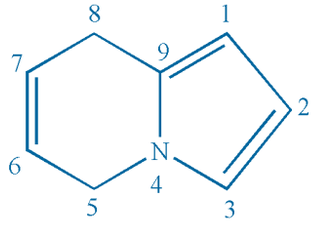
Acridine



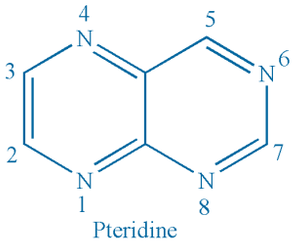
Indazole

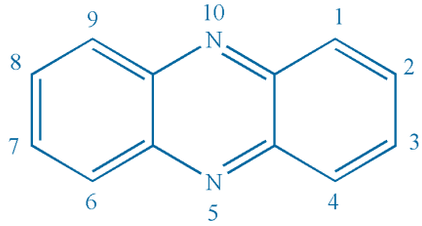


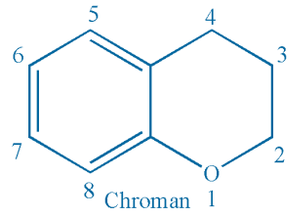
Benzimidazole

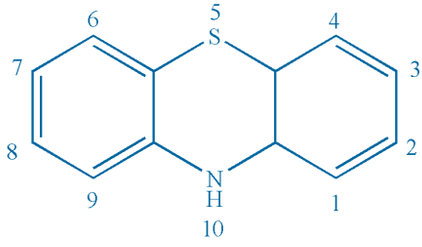


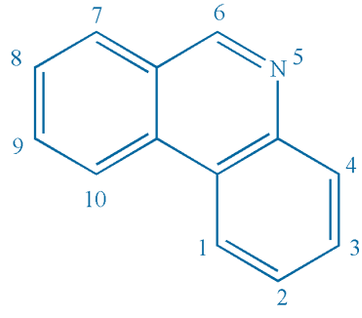
Indolizine


Pteridine


Phenazine


Chroman


Phenothiazine

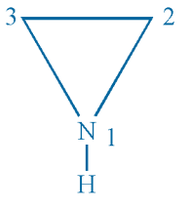

Phenanthridine

Classification of Heterocyclic Compounds

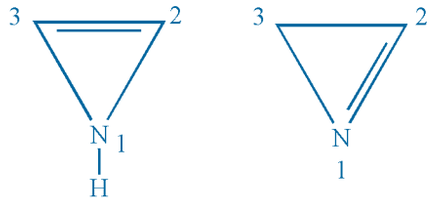
I. Three membered heterocyclic compounds with hetero atom.

(a) With nitrogen

Saturated Compound

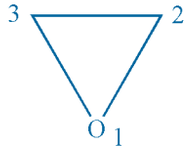

Aziridine

Unsaturated Compound

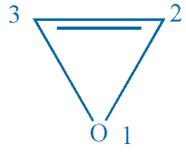

Azirine

(b) With oxygen

Saturated Compound

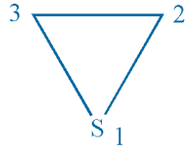

Oxirane

Unsaturated Compound

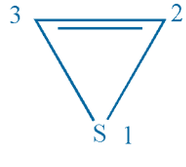

Oxirene

(c) With sulphur

Saturated Compound


Thiirane

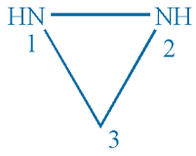
Unsaturated Compound


Thiirene

II. Three membered heterocyclic compounds with two hetero atoms.

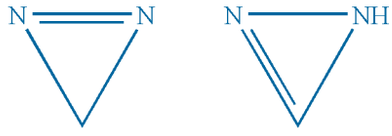
(a) With two nitrogen atoms

Saturated compound



Diaziridine

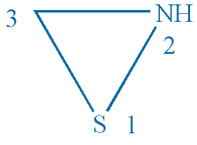
Unsaturated compounds



Diazirine

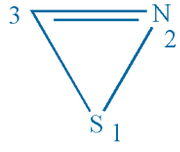
(b) With one nitrogen and one sulphur atoms:

Saturated compound



Thiaziridine

Unsaturated compound



Thiazirene

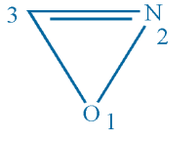
(c) With one nitrogen and one oxygen atoms:

Saturated compound



Oxaziridine

Unsaturated compound

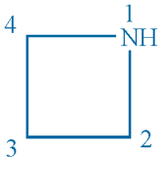


Oxaziren
e

III. Four membered heterocyclic compounds with one hetero atom.

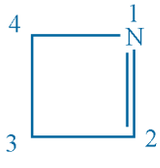
(a) With nitrogen

Saturated Compound

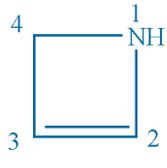


Azetidine

Partially Unsaturated Compounds

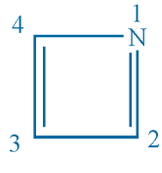


1-Azetine



2-Azetine

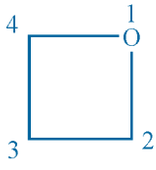
Completely Unsaturated Compound



Azete
(Antiaromatic)

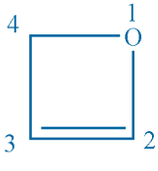
(b) With oxygen atom

Saturated Compound



Oxetane

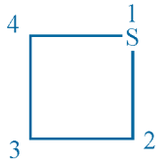
Unsaturated Compound



2-Oxetene

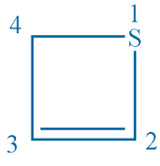
(c) With sulphur atom

Saturated Compound



Thietane

Unsaturated Compound

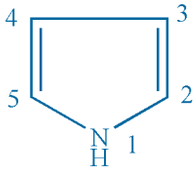


Thietene or Thiete

IV. Five membered heterocyclic compounds with one heteroatom.

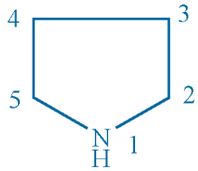
(a) With nitrogen

Unsaturated Compound



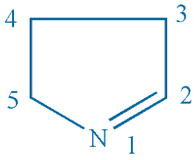
Pyrrole

Saturated Compound

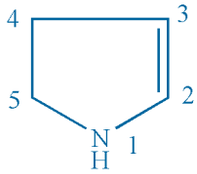


Pyrrolidine

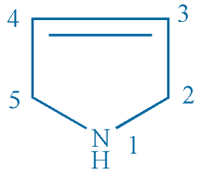
Partially Saturated Compounds



Δ^1-Pyrroline



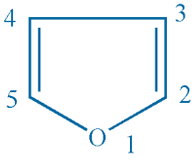
Δ^2-Pyrroline



Δ^3-Pyrroline

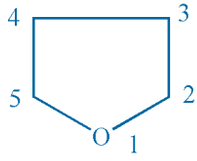
(b) With oxygen atom

Unsaturated Compound



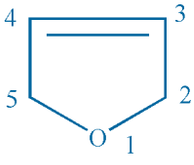
Furan

Saturated Compound

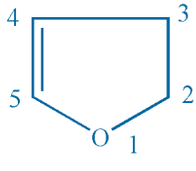


Tetrahydro furan

Partially Saturated Compounds



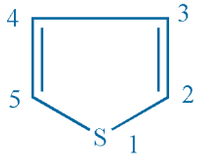
2,5-Dihydrofuran



2,3-Dihydrofuran

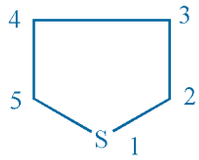
(c) With sulphur atom

Unsaturated Compound



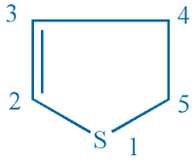
Thiophene

Fully Saturated Compound

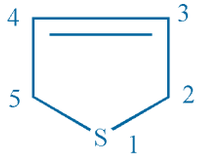


Tetrahydro thiophene (Thiolene)

Partially Saturated Compound



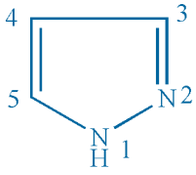
2-Thiolene
(2,3-Dihydrothiophene)



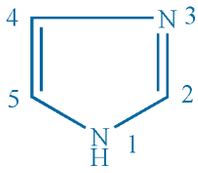
3-Thiolene
(3,4-Dihydrothiophene)

V. Five membered heterocyclic compounds with two hetero atoms.

(a) With two nitrogen atoms

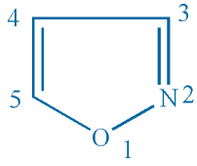


Pyrazole

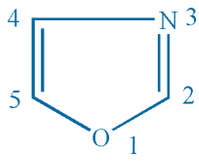


Imidazole

(b) With one nitrogen and one oxygen atom

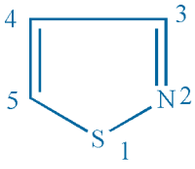


Isooxazole

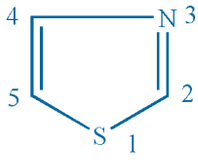


Oxazole

(c) With one nitrogen and one sulphur atom

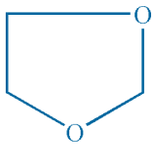


Isothiazole



Thiazole

(d) With two oxygen atoms

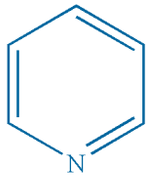


Dioxolane

VI. Six membered heterocyclic compounds with one hetero atom.

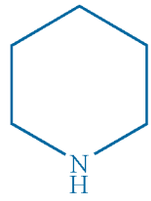
(a) With nitrogen atom

Unsaturated Compound



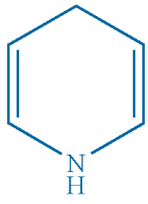
Pyridine

Fully Saturated Compound

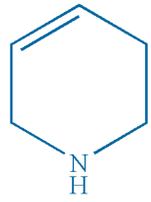


Piperidine

Partially Saturated Compounds

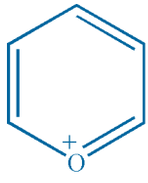


Dihydropyridine

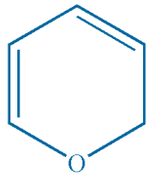


Tetrahydropyridine

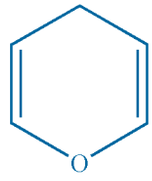
(b) With oxygen



Pyrylium salt



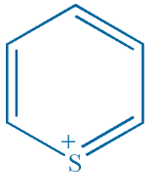
α-Pyrrones



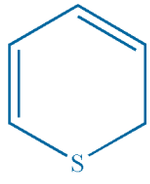
γ-Pyrrones

(Unstable and less aromatic)

(c) With sulphur



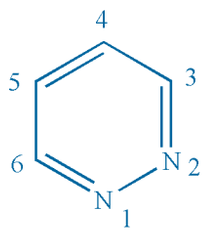
Thiopyrilium salt



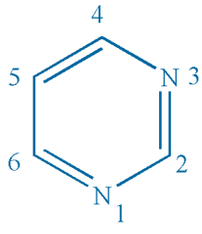
Thiopyran

VII. Six membered heterocyclic compounds with two hetero atoms.

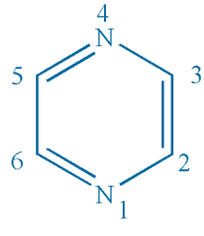
(a) With two nitrogen atoms



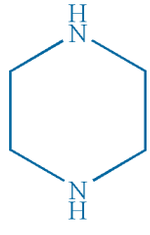
Pyridazine



Pyrimidine

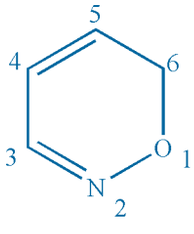


Pyrazine

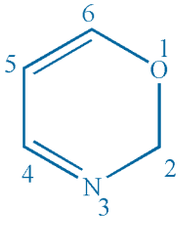


Piperazine

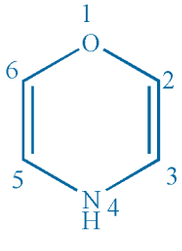
(b) With oxygen and nitrogen atoms



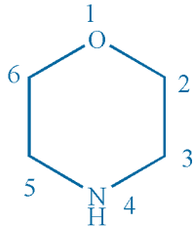
1,2-Oxazine



1,3-Oxazine

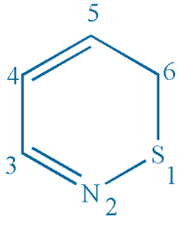


1,4-Oxazine

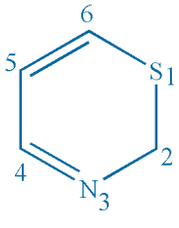


Morpholine

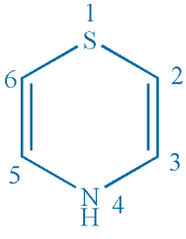
(c) With nitrogen and sulphur atoms



1,2-Thiazine

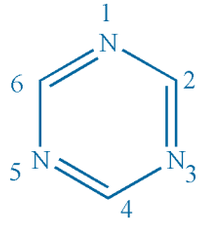


1,3-Thiazine



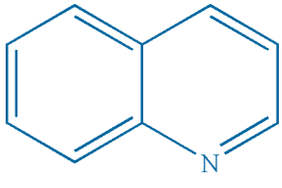
1,4-Thiazine

(d) With three nitrogen atoms

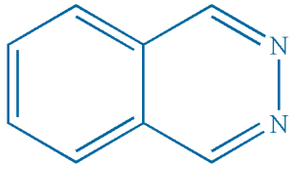


1,3,5-Triazine

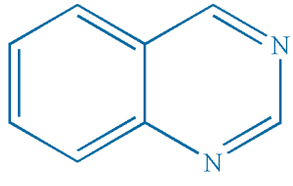
Fused rings



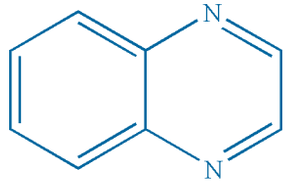
Quinoline



Pthalazine

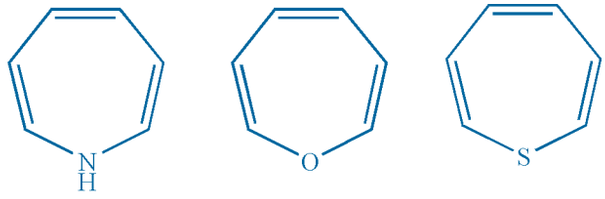


Quinazoline



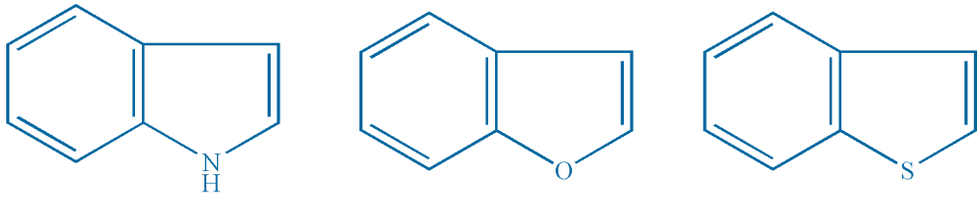
Quinoxaline

VIII. Seven membered heterocyclic compounds.

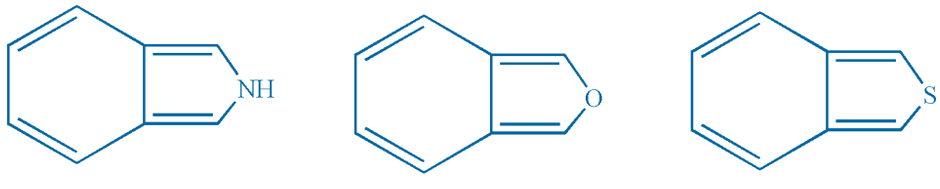


Azepine Oxepine Thiepine

IX. Bicyclic ring systems from pyrrole, furan and thiophene (fused five membered ring system).

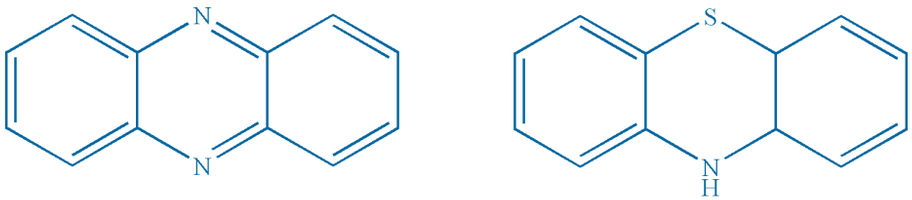


Indole Benzofuran Benzothiophene

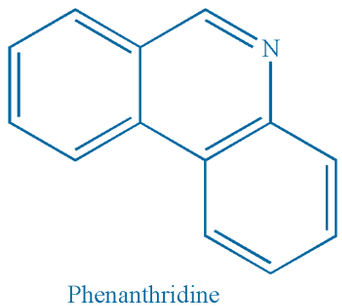


Isoindole Isobenzofuran Isobenzothiophene

X. Tricyclic heterocyclic compounds.

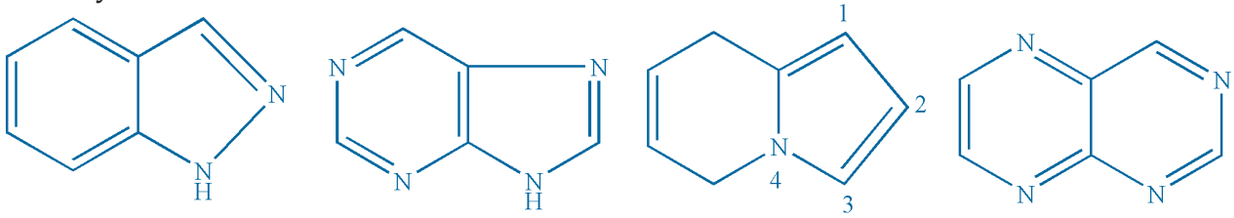


Phenazine Phenothiazine



Phenanthridine

Heterocycles with Trivial names.



Indazole Purine Indolizine Pteridine